Boron and Carbon Family

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p-Block Elements-1

Introduction to Group 13 Elements

Boron Family, Atomic and Physical properties Preparation of Boron, Chemical Properties of boron family

In periodic table there are four blocks.

(i) s-Block (ii) p-Block

(iii) d-Block (iv) f-Block

In this chapter we will discuss p-Block.

When electron arrives to the outer most p-orbital are known as p-Block elements and maximum number of electrons that can be accommodated in set of p-orbital is six, so according to this p-block classified into six groups in the periodic table.

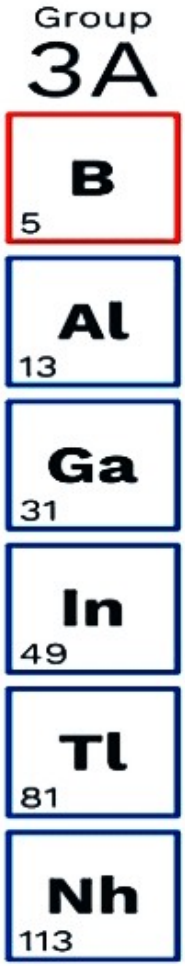
Classification of p-block-

(i) Boron family (ii) Carbon family

(iii) Nitrogen family (iv) Oxygen family

(v) Halogen family (vi) Nobel gases

But in this chapter, we only study about boron and carbon family.



(i) Boron Family

It is a present in group 13 in periodic table and it contain boron (B), aluminium (Al), gallium (Ga), indium (In) and Thallium (Tl).

They all collectively known as boron family but sometimes it calls as aluminium family.

Note:

In group 13 elements, boron is the only non-metal of this group remaining other elements are metals.

Boron shows non-metallic nature because it has high ionisation energy and its small size.

Basic properties of Boron family :

1. **Electronic Configuration**

ns^2np^1 is the general outer electronic configuration.

Hence these elements belong to p-block.

Elements	Configuration
Boron [$_5$B]	[He]$2s^2 2p^1$
Aluminium [$_{13}$Al]	[Ne]$3s^2 3p^1$
Gallium [$_{31}$Ga]	[Ar]$3d^{10} 4s^2 4p^1$
Indium [$_{49}$In]	[Kr]$4d^{10} 5s^2\ 5p^1$
Thallium [$_{81}$Tl]	[Xe]$4f^{14} 5d^{10}\ 6s^2 6p^1$

2. Occurrence

Boron is fairly rare element. It mainly occurs as **orthoboric acid (H_3BO_3), Colemanite ($Ca_2B_6O_{11}.5H_2O$), Borax (Tincal) ($Na_2B_4O_7.10H_2O$), Boronatro calcite ($CaB_4O_7.NaBO_2.8H_2O$), Boracite ($2Mg_3B_8O_{15}.MgCl_2$), Kernite ($Na_2B_4O_7.4H_2O$)**. In India borax found in puga valley (Ladakh), sambhar lake (Rajasthan).

Natural Abundance of boron = 0.0001%

Boron have two isotopes

B^{10} (19%) B^{11} (81%)

In earth curst aluminium is 3rd most abundant element (8.3% by mass) and aluminium is most abundant metal.

Note: O > Si > Al > Fe (Abundance order of elements in atmosphere).

Aluminium found in many forms but mostly found in corundum (Al_2O_3), diaspore ($Al_2O_3.H_2O$),Bauxite($Al_2O_3.2H_2O$) & cryolite (Na_3AlF_6)

Ga, In & Tn are less abundant.

B is nonmetal but Al, Ga, In and Tl show typical metallic properties.

Tl is highly toxic.

Pysical properties of boron family :

When we go down in the group, we found regular increasing trend in density.

In case of melting point first it reduces than increase from B to Ga it reduce then increase.

Boron exists as giant covalent polymer in both solid and liquid state this is happened because it has very high melting point.

Gallium (Ga) has low melting point (29.8oC) because only Ga2 molecule it exists as liquid in summer. It also exists as liquid up to 2000oC this is because it used in high temperature thermometry.

In case of boiling point these elements show regular drop down in the group.

B-non-metal, black colour, solid B_{12}, Icosahedral, high M.P. Other members are soft metals, with high M.P. and high electrical conductivity.

B_2 does not exist. It exist as B_{12}.

EXERCISE:

Q1 **Which of the subsequent are the ores of Boron?**
(A) Tincal **(B) Kernite** **(C) Colemanite** **(D) Bauxite**

Sol. **(ABC)**

Borax or Tincal: $Na_2B_4O_7.10H_2O$

Kernite: $Na_2B_4O_7.4H_2O$

Colemanite: $Ca_2B_6O_{11}$

Bauxite: $Al_2O_3.2H_2O$

Atomic and Physical properties

(i) Atomic and Ionic radii down the group decreases.

B < Al > Ga < In < Tl

Or

B < Ga < Al < In < Tl

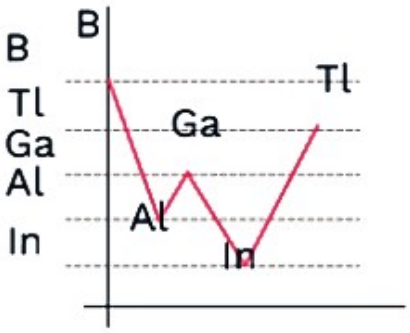
Al $\Rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$

Ga $\Rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1\ 3d^{10}\ 4s^2\ 4p^1$

Note:

Unexpected decrease in atomic radii of Ga is due to poor screening effect of electrons in d-subshell. Which causes more attraction (effective nuclear charge) towards outermost electron.

(ii) Ionization Enthalpies.



B > Al < Ga > In < Tl (Sum of three IE values)

Or

B > Tl > Ga > Al > In

$Ga \Rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^1$

$Tl \Rightarrow [Xe]4f^{14}\ 5d^{10}\ 6s^2\ 6p^1$

Note:

In case of gallium and thallium observed irregularity. The reason of this irregularity is ineffective shielding of nuclear charge by 3d(Ga) and 4f, 5d (Tl) subshell respectively.

(iii) Melting and Boiling points

M.P.: B > Al > Tl> In > Ga

B.P.: B > Al > Ga > In > Tl

(iv) Electropositive Character

Due to high I, E they are less electropositive. On moving down the group, metallic character raises due to decrease in IE.

[∴ B is nonmetal and other elements are metals.

$$\frac{B}{\text{Non metal}} < \frac{Al < Ga < In < Tl}{\text{Metal}}$$

(v) Density

When we go down in the group, we found regular increasing trend in density.

(vi) Oxidation potential

(a) These elements have high standard oxidation potentials.

	B	Al	Ga	In	Tl
E^0 for $M \rightarrow M^{3+} + 3e^-$	–	+1.66	+0.56	+0.34	+1.26
E^0 for $M \rightarrow M^+ + e^-$	–	+0.55	–	+0.18	+0.34

(b) Boron has very low oxidation potentials, so boron does not form positive ions in aqueous solution.

(c) Due to higher heat of hydration on account of smaller size of trivalent cations, their values of standard oxidation potential are high.

Note:

We use carbon as reducing agent but when we compare Al and C, aluminium is a strong reducing agent and can reduce oxides.
This is happening because, Aluminium has lower ionisation energy than carbon.

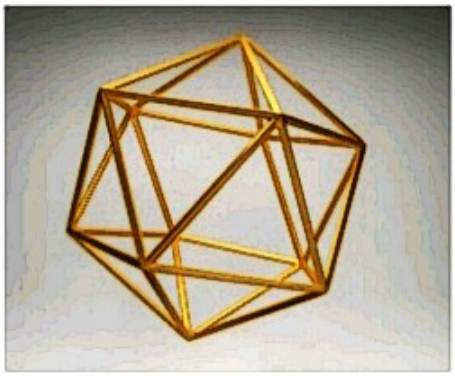
(vii) Complex formation:

They tendency to form more complexes than s-block elements this is happens because they have vacant orbitals to accept electrons, and their smaller size and high effective nuclear charge.



Note:

Boron exists in many allotropic forms. All the allotropes have basic building B_{12} icosahedral units made up of polyhedron having 20 faces and 12 corners. For example, one is the simplest form: α-rhombohedral boron. But Al, In and Tl all have close packed metal structure.



(viii) Oxidation sates:

Generally higher oxidation state is more stable but in p-block 13 to 15 group on moving down the group lower oxidation state becomes more stable. This is for the reason that of poor shielding effect of 4f sub shell (inert pair effect).

Element	O. S		Stable O. S
B		+3	+3
Al	+1	+3	+3

Ga	+1	+3	+3
In	+1	+3	+3
Tl	+1	+3	+1

If we ignored inert pair effect and we go down in the group 13, the trend to achive +3 oxidation state goes on reduce and the tendency to gain + 1.
Order of stability of + 1 oxidation state is Ga < In < Tl

Compounds have Tl+ is more stable than compounds have Tl^{+3}.

Due to good shielding of 4s, subshell 6p electron is less attracted towards nucleus and it easy to come out 6p electron hence (+1) O.S. is more stable.

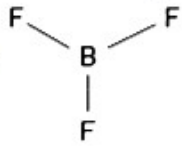
Lower the energy required to take out the electron more will be stability.

Stability of ion ($Tl^{+} > Tl^{+3}$) ($Tl^{3+}$is an oxidizing agent)

The tendency of inertness of electron towards bonding is called inert pair effect.

$B = 1s^2\ 2s^2\ 2p^1$

B^{+3} does not form due to the sum of three ionization energy is very high. Because of very much higher ionization energy B^{+3} is unstable.

e.g.  is covalent compound. Here B can show +3 formal charge,

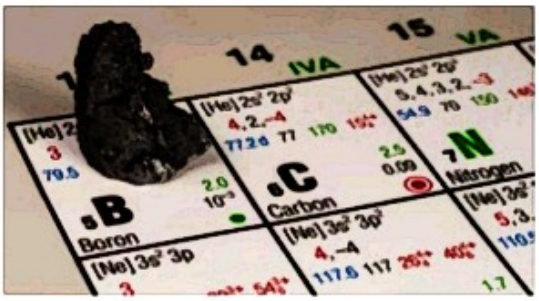
but +3 ation does not form.

Q2 **Which among the following is most basic and why?**
Al_2O_3, Tl_2O_3, Tl_2O

Sol.

	Oxidation state	
Al_2O_3	Al = +3	Ionization energy decreases
Tl_2O_3	Tl = +3	Basic nature increases
Tl_2O	Tl = +1	

Isolation of Boron:



By the reduction of boric oxide with highly electropositive metals like Mg, K, Na, Al, etc. we acquired dark brown power. in the lack of boron halides and air with hydrogen at high temperature..

For example

$$B_2O_3 + 6K \xrightarrow{\text{heat}} 2B + 3K_2O$$

$$2BCl_3 + 3H_2 \xrightarrow{1270K} 2B + 6HCl.$$

Through thermal decomposition of boron triiodide over red hot tungsten (W) filament and boron hydrides. For example,

$$2BI_3 \xrightarrow{\text{W Heat}} 2B + 3I_2$$

$$B_2H_6 \xrightarrow{\text{heat}} 2B + 3H_2$$

Preparation of Boron:

(1) From Borax:

$$Na_2B_4O_7 + \text{conc.}\,H_2SO_4 \longrightarrow NaHSO_4 + H_2B_4O_7$$

$$H_2B_4O_7 + H_2O \longrightarrow H_3BO_3$$

$$Na_2B_4O_7 + \text{conc.}\,H_2SO_4 \longrightarrow H_3BO_3 + NaHSO_4$$

(2) From Boric acid:

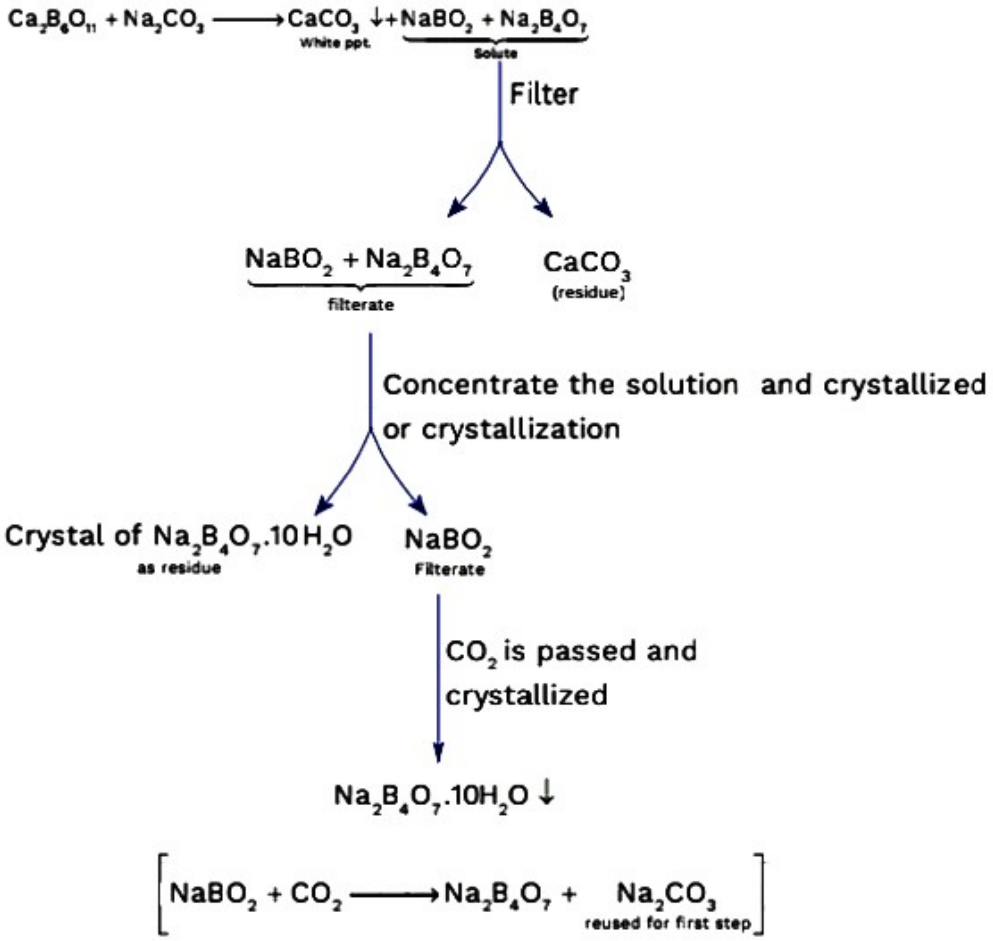
$$H_3BO_3 \xrightarrow{100^\circ C} HBO_2 \text{ (metaboric acid)}$$

$$HBO_2 \xrightarrow{140^\circ C} H_2B_4O_7 \text{ (tetraboric acid or pyroboric acid)}$$

$$H_2B_4O_7 \xrightarrow[\text{temperature}]{\text{high}} B_2O_3 + H_2O$$

$$B_2O_3 + Li / Na / K / Al \longrightarrow B + Li_2O / Al_2O / K_2O / Al_2O_3$$

(3) From Colemanite



Chemical Properties:

Reactivity towards air

(i) At normal temperature pure form of Boron is almost unreactive when it heated it react with air and form Boric oxide (B_2O_3). In case of water, it does not react.
Aluminium give Al_2O_3 when it burns in air

(ii) When heated gallium and indium it reacted towards air otherwise, they are not affected by air. Tl is little more reactive towards air, and it form an oxide film at surface. It moisture air, a layer of Tl(OH) is produced.

(iii) Aluminium have tendency to decomposes H2O and react easily in air at normal temperature to from a protecting film of its oxide. Which protects it.

Oxides and hydroxides

(i) Members of Boron family form hydroxides and oxides and their general formulas are $M(OH)_3$ and M_2O_3 respectively.

(ii) Nature of Hydroxide = Acidic nature
Nature of oxide = from B to Tl it acidic to basic through amphoteric.

B_2O_3 and $B(OH)_3$ > Al_2O_3 and $Al(OH)_3$ >
(acidic) (amphoteric)
Ga_2O_3 and $Ga(OH)_3$ > In_2O_3 and $In(OH)_3$ > Tl_2O_3 and $Tl(OH)_3$
(amphoteric) (basic) (strong basic)

$B(OH)_3$ or H_3BO_3 is weak monobasic Lewis acid

(iii) Except boric acid all hydroxide of group 13 are insoluble in water and form a gelatinous precipitate.
In case of Boric acid, it soluble in water it accept lone pair and behaves as Lewis acid.

$B(OH)_3 + H_2O \rightarrow B(OH)_4^- + H^+$

(iv) Oxides of aluminium are amphoteric in nature they dissolve in acid as well as is base.

$$Al_2O_3 + 3H_2SO_4 \longrightarrow Al_2(SO_4)_3 + 3H_2O$$

$$Al_2O_3 + 2NaOH \xrightarrow{\text{fuse}} \underset{\text{Sodium meta aluminate}}{2NaAlO_2} + H_2O$$

(v) Corundum is a crystalline form of Al_2O_3 and it formed when we heating amorphous form of Al_2O_3 to 2000K.
Corundum is very tough and used as abrasive.

Action of Alkalies

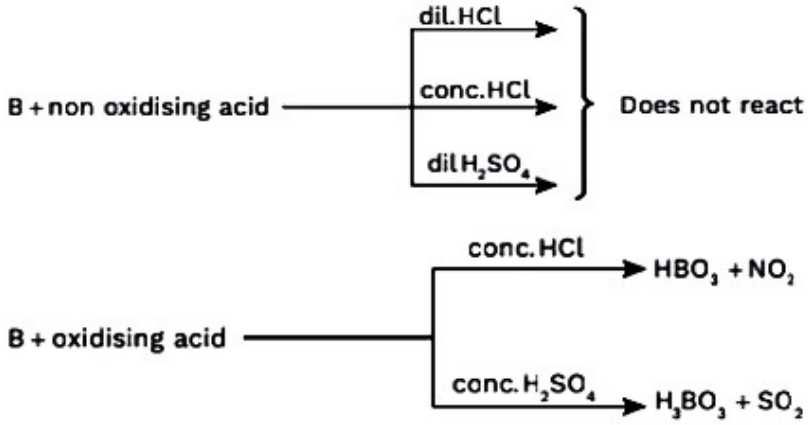
(i) Group 13 element boron only dissolves in fused bases.
$2B + 6NaOH \text{ (fused)} \rightarrow 2Na_3BO_3 + 3H_2$

(ii) But in case of Ga and Al dissolves in fused as well as aqueous bases.
$2Al + 2NaOH + 2H_2O \rightarrow 2NaAlO_2 + 3H_2$

(iii) In is unaffected towards bases even on heating.

(vi) B > Al > Ga > In ⇒ Decreasing order of Lewis acid nature of Halides of Boron family.

Actions of Acids



Q3 **Aqueous solution of $AlCl_3$ behaves acidic towards litmus while that of NaCl does not. Why ?**

Sol. $Al_2Cl_3 + 6H_2O \rightarrow 2Al(OH)_3 + HCl$ due to presence of HCl, the aq. solution is acidic toward litmus. NaCl is ionic and is dissociated in ions.

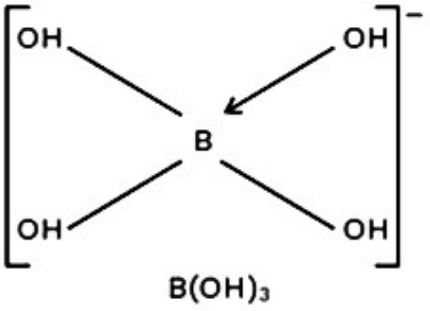
Q4 **$B(OH)_3$ is an acid, $Al(OH)_3$ is amphoteric and $Tl(OH)$ is basic. Why?**

Sol. $H_3BO_3 + H_2O \rightarrow H_3O^+ + H_2BO_3^-$

Weak acid monobasic acid

Orthoboric Acid doesn't donate Proton, but rather it accept OH^-, therefore it is a lewis acid.

→ $Al(OH)_3$ it reacts Both acid as well as base thus it behave like Amphoteric.



$B(OH)_3$

→ $Al(OH)_3$ it reacts Both acid as well as base thus it behave like amphoteric.

B Al Ga In Tl

Ionisation energy decreases

→ Basic nature increases due to easily donate electron.

Thus $Tl(OH)_3$ is Basic.

Q5 **Boron shows amphoteric nature. Illustrate with chemical reaction.**

Sol. Boron react with Acid as well as Base So, behave as amphoteric nature.

$B + 3HNO_3 \rightarrow H_3BO_3 + 3NO_2\uparrow$

$B + 6NaOH \rightarrow 2Na_3BO_3 + 3H_2\uparrow$

Reaction with water:

$B + H_2O \xrightarrow[\text{Hot/cold}]{} \text{No reaction}$

$B + H_2O \xrightarrow[\text{Red hot}]{} B_2O_3 + H_2$

Reaction with metal:

Many metals form borides with boron. These are often non-spoichiometric.

B + Metal → Cr → CrB; Mn → Mn_3B_2

Reaction with nonmetals:

B + Non-Metal → O_2 → B_2O_3; C → B_4C (Carbides: used to make bullet proof jackets); N → $(BN)_x$ (Inorganic benzene); S → B_2S_3; X_2 → BX_3

EXERCISES:

Q6 **Find out the correct statements?**
(A) All III A group elements form trihalide
(B) Only Al reacts directly with N_2 at high temperature and forms AlN
(C) Al is passivated by concentrated HNO_3
(D) All IV A group elements form tetrahalides

Sol. **(ABC)**

All 13th group elements forms trihalide even Tl^+ as TlI_3, BX_3, $AlCl_3$, $GaCl_3$, $InCl_3$

Q7 **Boron tribromide is stronger acid than boron trifluoride. Explain.**

Sol. The tendency of Back bonding is maximum in BF_3 and decreases from BF_3 to BI_3 as the size of Halogen increases it's p orbital gain high energy shell's in bromine overlap less effectively with vacant p-orbital of boron in BBr_3.

Thus, BBr_3 is stronger Acid than BF_3.

Q8 **The pπ–pπ back bonding occurs in the halides of boron but not in the halides of aluminium. Why?**

Sol. The tendency to show pπ-pπ back bonding depends on the size of central atom. This tendency decreases as the size of central atom increases since Al has longer size than boron. The back bonding is not possible.

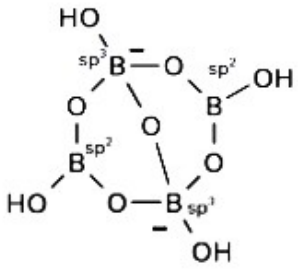
Q9 **$AlCl_3$ forms a dimer but BCl_3 does not form dimer. Why?**

Sol.

$AlCl_3$	BCl_3
3p – 3p (shell)	2p – 3p(shell)
Al-because large size Aluminimum metal atoms complete their octet by co-ordinate bond forming bridges by Cl atom between two Al atom.	due to small size of B it can't form co-ordinate bond with Cl atom. Thus dimer is not possible.

Compounds of Boron, Anomalous Behaviour of Boron, Diagonal relationship between Boron and Silicon, Uses of boron, Compounds of Aluminium

Compounds of Boron



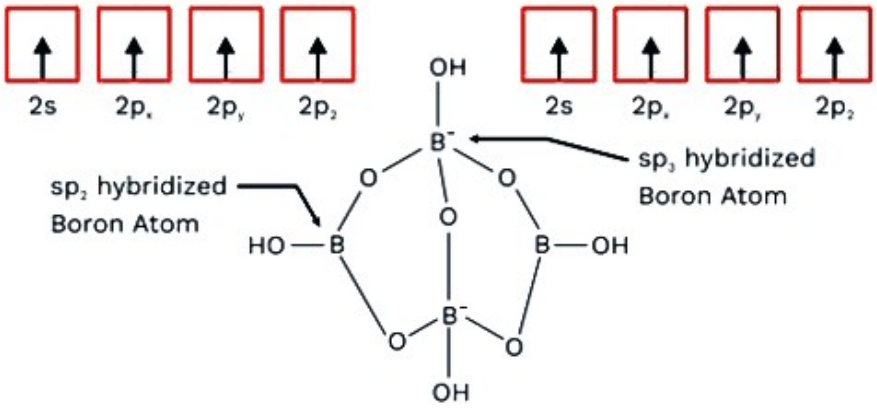
1. Borax: $Na_2B_4O_7.10H_2O$ or $Na_2[B_4O_5(OH)_4].8H_2O$

(1) It has two **tetrahedral** unit (sp^3).

(2) It has two **trigonal planar** unit (sp^2).

(3) All boron atoms are not in same plane.

The number of B–O–B linkage is 5.



Structure of Boron

It happens naturally as **tincal** (Suhaga) which contains about 50% borax in certain lakes. It is also achieved from the mineral colemanite by boiling it with a solution of Na_2CO_3.

$$\underset{\text{Colemanite}}{Ca_2B_6O_{11}} + 2Na_2CO_3 \longrightarrow \underset{\text{Borax}}{Na_2B_4O_7} + 2CaCO_3 + 2NaBO_2$$

Properties:

(a) Its aqueous solution **is basic** due to hydrolysis.

$$Na_2B_4O_7 + 7H_2O \longrightarrow \underbrace{2NaOH + 4H_3BO_3}_{\substack{(1)\text{ basic solution} \\ (2)\text{buffer solution}}}$$

(b) When borax is heated it used its water of crystallization and convert into fluffy mass and further heating, it melts and convert into clear liquid which solidifies to a transparent glassy bead contains of sodium metaborate ($NaBO_2$) and boric anhydride (B_2O_3).

$$Na_2B_4O_7.10H_2O \xrightarrow[-10H_2O]{\Delta} Na_2B_4O_7 \xrightarrow[\text{Borax bead}]{\Delta} 2NaBO_2 + B_2O_3$$

Borax bead test is basically, where borax bead is utilized for the exposure of coloured basic radicals.

Borax Bead Test:

Basically, it is a mixture of $NaBO_2$ and $B_2O_3.B_2O_3$ on heating this mixture it easily with a number of coloured transition metal oxides such as Cr, Cu, Co, Ni, Mn etc. to form the corresponding metaborates which possess characteristic colours.

$$CoSO_4 \xrightarrow{\Delta} CoO + SO_3; \quad CoO + B_2O_3 \longrightarrow \underset{\text{Cobalt metaborate (Blue)}}{Co(BO_2)_2}$$

Colours of some important metaborates are:	Cupric metaborate, $Cu(BO_2)_2$: dark blue
	Chromium metaborate, $Cr(BO_2)_2$: green
	Nickel metaborate, $Ni(BO_2)_2$: brown
	Manganese metaborate, $Mn(BO_2)_2$: pink violet

(c) When we heated with C_2H_5OH and conc. H_2SO_4. It gives us tri ethyl borate as volatile vapours which burns with green edged flame.

$$Na_2B_4O_7 + H_2SO_4 + 5H_2O \longrightarrow Na_2SO_4 + 4H_3BO_3$$

$$H_3BO_3 + 3C_2H_5OH \longrightarrow \underset{\text{Triethyl borate}}{B(OC_2H_5)_3} + 3H_2O$$

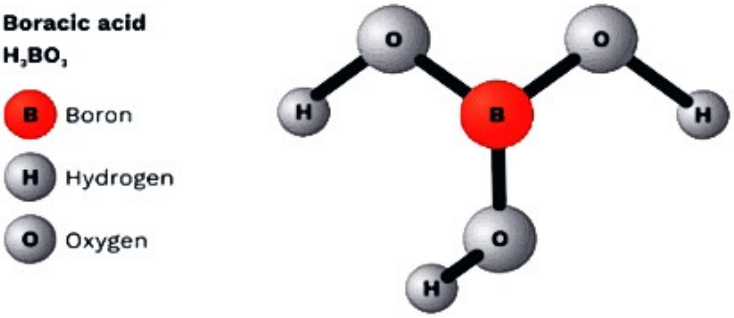
Its qualitative analysis, for borate radicals test we use this reaction.

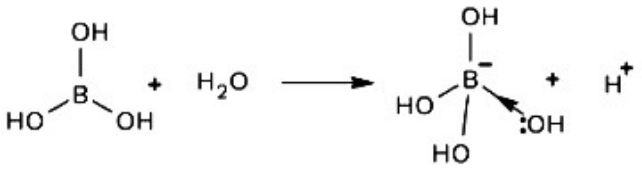
Uses: (a) We use this for creating hard and optical glasses.
(b) In lab we use for borax bead test.
(c) Softening of water.
(d) It has antiseptic character, so we use it for formation of medicals sops.

2. Boric acid (H_3BO_3)

1. H_3BO_3 is white crystalline solid, soapy in touch. It is soluble in water.
2. It is weak mono basic Lewis acid.
3. It does not act as a proton donor acid.

Boracic acid
H_3BO_3





Or

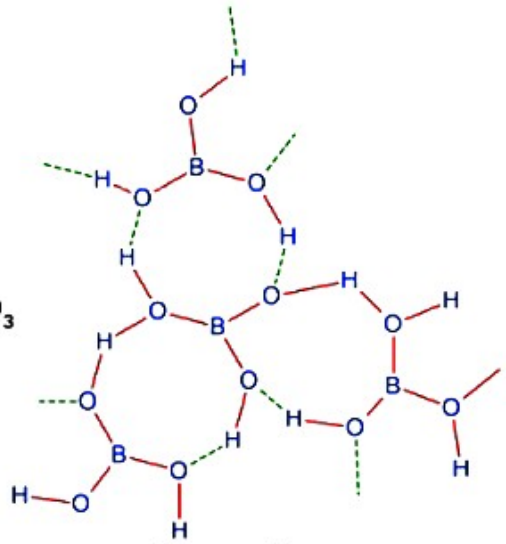
$$B(OH)_3 + H_2O \rightleftharpoons [B(OH)_4^-] + H^+$$

Borate ion

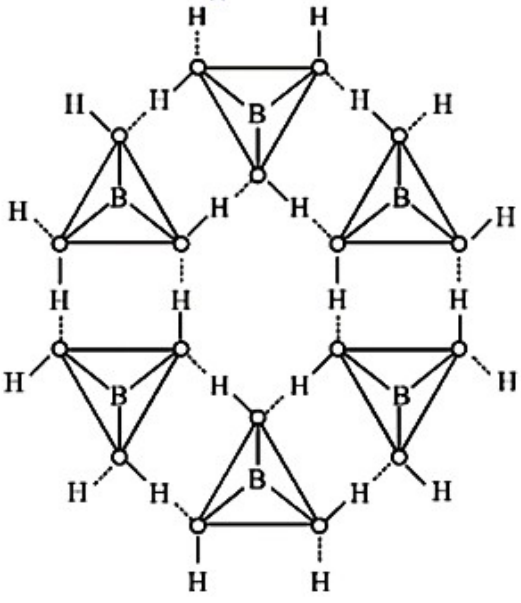
When heated if first forms metaboric acid (HBO_2) and then boron trioxide.

$$\underset{\text{Boric acid}}{H_3BO_3} \xrightarrow{100°C} HBO_2 \xrightarrow{160°C} H_2B_4O_7 \xrightarrow{\text{red hot}} B_2O_3$$

Orthoboric acid is soapy to touch. It is less soluble in cold water (H_2O) but more solvable in hot water. In the solid-state, the $B(OH)_3$ units are hydrogen bonded all together in two dimensional sheets with almost hexagonal symmetry.



Boric acid structure and in this structure the dotted lines show hydrogen bonds.



Q10 **Why $B(OH)_3$ does not give titration with NaOH?**

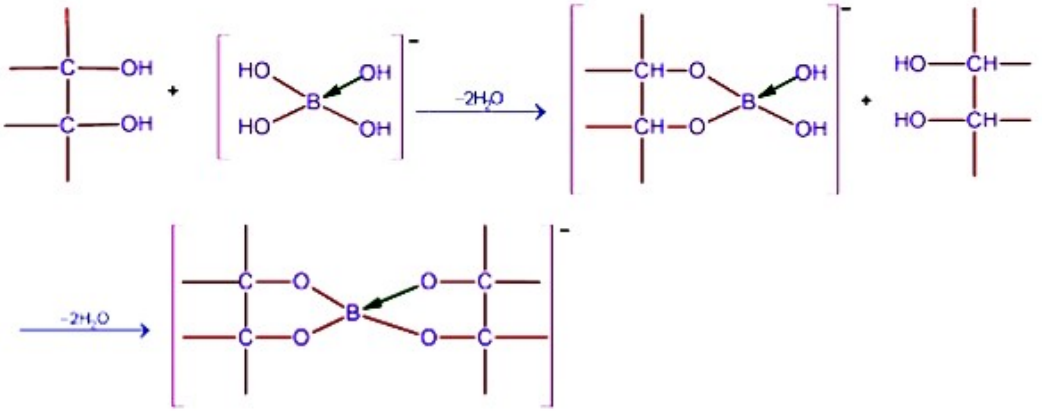
Sol. **(B)**

$B(OH)_3$ does not give titration with NaOH because a sharp end point is not obtained but in presence of polyhydroxy organic acid, it acts as strong acid and give titration with NaOH.

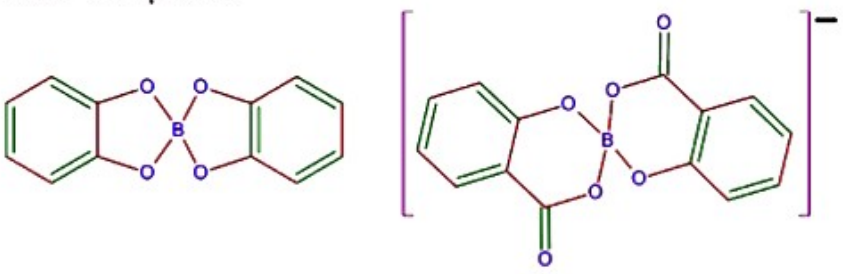
$$B(OH)_3 + NaOH \rightleftharpoons Na[B(OH)_4] \rightleftharpoons NaBO_2 + 2H_2O$$

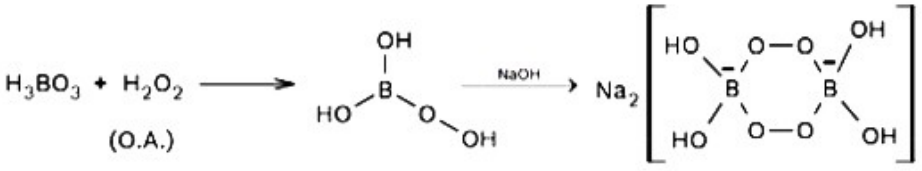
Sodium metaborate

$B(OH)_3X$ as a strong acid in presence of the cis–diol except ethylene glycol.



Ethanol does not form related complex but catechol, salicylic acids, mannitiol form similar complexes





Sodium peroxidiborate

It is used as brightner in washing powder.

Use:

1. It is an antiseptic & its water solution is used as an eyewash.
2. It is similarly used in glass, enamel and pottery industry.

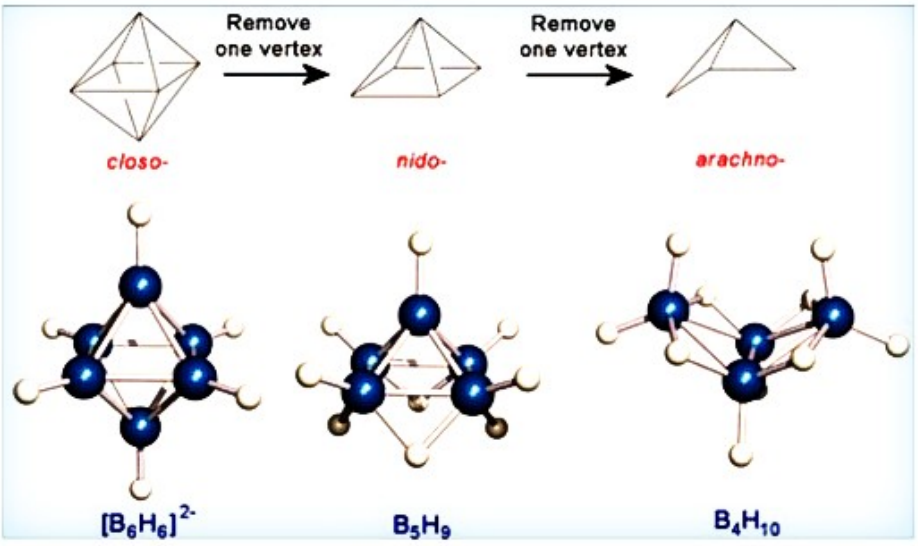
3. Diborane:

Boron forms hydrides of the types B_nH_{n+4} and B_nH_{n+6} known as boranes.

Diborane is the easiest boron hydride which is a dimer of BH_3.

Borane

Nido borane	Arachnoborane
B_nH_{n+4}	B_nH_{n+6}
B_2H_6 (Diborane)	B_2H_8
B_4H_8	B_4H_{10}



Preparation of Diborane

$$LiAlH_4 + BF_3 \longrightarrow LiF + AlF_3 + B_2H_6$$

$$8BF_3 + 6LiH \xrightarrow{450K} B_2H_6 + 6LiBF_4$$

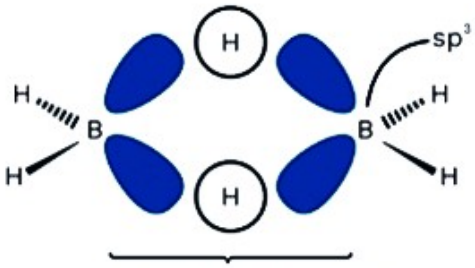
Lab method:

In the laboratory, it is made by the oxidation of sodium borohydride with I_2.

$$2NaBH_4 + I_2 \longrightarrow B_2H_6 + 2NaI + H_2$$

Industrial method:

$$2BF_3 + 6NaH \xrightarrow{450K} B_2H_6 + 6NaF$$



2e⁻ spread over 3 orbitals

3-Center, 2-Electron Bond

Properties:

i. B_2H_6 is colourless, highly toxic gas, with a boiling point of 180 K.

ii. Diborane catches fire spontaneously upon exposure to air.

$$B_2H_6 + 3O_2 \longrightarrow 2B_2O_3 + 3H_2O;\ \Delta H = -1976KJ/mole$$

iii. $B_2H_6 + \underset{(Cold)}{H_2O} \longrightarrow B(OH)_3 + 6H_2\uparrow$

$$B_2H_6 + X_2 \longrightarrow B_2H_5X + HX$$

$$B_2H_6 + HX \longrightarrow B_2H_5X + H_2$$

iv. In organic synthesis both $LiBH_4$ and $NaBH_4$ both are used as reducing agent.

$$B_2H_6 + MH \longrightarrow \underset{M=Li\ or\ Na}{M^+(BH_4)^-}$$

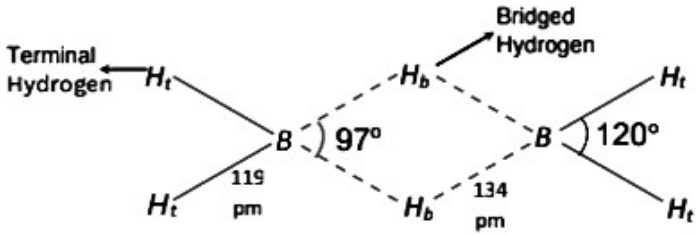
v. All Boranes are known as electron-deficient compounds or Lewis acids because boron in boranes never complete it's octate.

Structure of diborane:

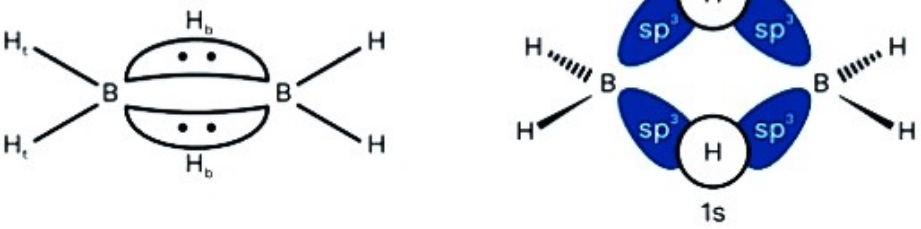
Diborane (B_2H_6) has $3C-2e^-$ bond also known as banana shape bond.

(a) $B-H_t$ → It is two center electron pair bond → $3C-2e^-$
It is a normal covalent bond.

(b) $B-H_b$ → This bond is made between three atoms $B-H_b-B$ that's why it is known as $3C-2e^-$ pair bond.



OR



Structure of diborane (B_2H_6)

4. Boron Halides

Boron reacts with halogens on great heating to form boron halides.

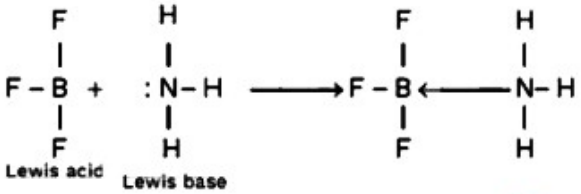
$$2B + 3X_2 \xrightarrow{\text{Heat}} 2BX_3 \ (X = F, Cl, Br, I)$$

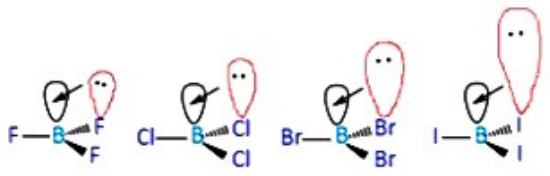
Note: **BF_3,** **BCl_3 and BBr_3,** **BI_3**

↓ ↓ ↓

Gaseous state Liquid state Solid state

Boron halides are electron deficient compounds, and they act as a Lewis acid because in Boron halide boron is a central atom and has 3 shared pairs of electron with halogen atoms, so there are 2 electron less than the octet.





(2pπ—2pπ) (2pπ—3pπ) (2pπ—4pπ) (2pπ—5pπ)

Boron halides act as very effective lewis's acid because in BX_3, boron atom has 6 electrons in it's outer most orbit and thus it can accept a pair of electrons from gives it can accept a pair of electrons from gives molecules like NH3 to complete its octate.

Lewis activity order of boron trihalides : $BI_3 > BBr_3 > BCl_3 > BF_3$

If we talk about the general order, we found that order is just the reverse. In general order basis on the electronegativity of the halogen fluorine being the most electronegative element and it create the most electron insufficiency on boron and thus B in BF_3, easily accept electron pair from giver, but this is not correct.

This unusual behaviour has been explained by the tendency of the halogen atom to back donate its unused electrons to the vacant p orbitals of boron atom.

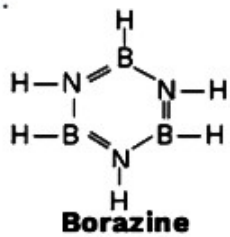
In boron tri halides, boron has a vacant 2p orbitals and each fluoride has totally filled unused 2p orbitals and these orbitals belongs to same energy level (2p) they can overlap effectively, and they formed P–P bond. This type or Bond formation known as back bonding or back donation. After that in bond B–F few double bonds character will be observed.

5. Borasole or Triborine triamine ($B_3N_3H_6$) or Borazine

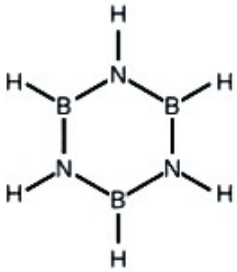
It is a combination of B, N and H. It is a colourless solution and is also known as inorganic benzene.

$$2B_2H_6 + 6NH_3 \xrightarrow{180^\circ C} 2B_3N_3H_6 + 12H_2 .$$

It has a 6 membered ring of alternating B and N atoms, each is further linked to a H–atom.



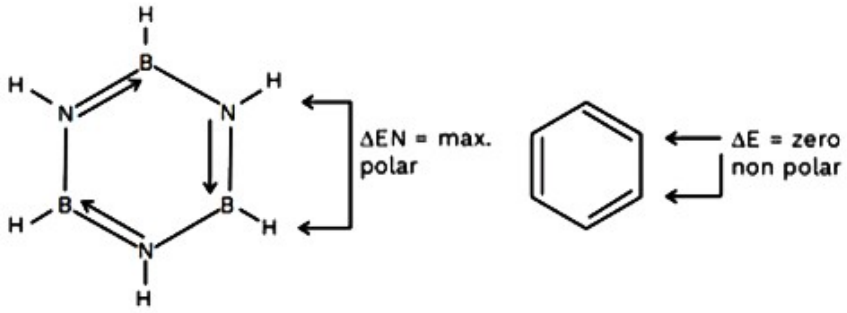
Borazine



Borazine

Q11 **Why Borazine is more reactive than benzene?**

Sol. In benzene C = C bond are non polar while in borozene the bond between boron and nitrogen are polar addition is frequent in borazine. The more negative group generally attached to boron atoms.



Borazole

6. Boron nitride (BN)

It is made by treating BCl_3 with an excess of NH_3 and pyrolyzing the resulting mixture in an atmosphere of NH_3 at 750°C.

$$BCl_3 + NH_3 \longrightarrow [H_3N \longrightarrow BCl_3] \xrightarrow[\text{excess } NH_3]{750^\circ C} BN + 3HCl$$

It is a **good insulator, colourless, diamagnetic and almost unreactive solid.**

Uses of Boron:

1. In atomic reactors as protective shields and control rods we use boron.
2. We use boron as a semiconductor for making electronic gadgets.
3. For hardness of steel.
4. Formation of light materials for aircrafts.

Anomalous Behaviour of Boron

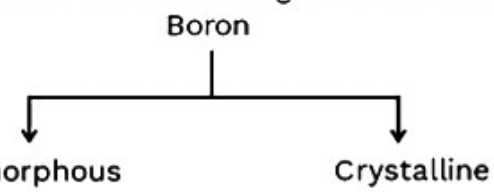
Like Be and Li, boron also show anomalous behaviour this is because it have extra small size and high nuclear charge/size ratio.

The important points about its anomalous behaviors :

(i) In group 13 elements only, boron is a non-metal other elements are metals.

(ii) In group 13 elements only, boron is a bad conductor of electricity while other metals are great conductors.

(iii)



Boron shows allotropy and exist into 2 forms and others does not exist in several forms.

(iv) Melting point (M.P.) and boiling point (B.P.) of boron are much greater like other non-metals.

(v) In group 13 elements boron only formed covalent compounds while aluminium and other elements formed some ionic compounds.

(vi) Boron oxide and hydroxide are acidic in nature while other elements oxide and hydroxide are basic and amphoteric in nature.

(vii) BX_3 survive as monomers and other side aluminium halides survive as dimers.

(viii) Boranes are stable whereas Al hydrides are unstable.

(ix) When boron reacts with diluted acid no action observed but in case of others they liberate hydrogen gas from them.

(x) Aluminates are more stables than borates.

(xi) There is difference in covalency, boron shows maximum 4 Example: BH_4^- while other members show maximum 6 Example: $[Al(OH)_6]^{3-}$.

(xii) Other members except boron decompose steam but boron does not.

(xiii) Except boron other members of group 13 from alloys while boron combines with metals from borides.

Example: Mg_3B_2.

(xiv) In case of boron when we treated boron with concentrated nitric acid, boron oxidises into boric acid, while no such action is noticed with other group representatives.

$B + 3HNO_3 \longrightarrow H_3BO_3 + 3NO_2$

Diagonal relationship between Silicon and Boron

As we known about the boron, boron differs from other elements of boron family because it has small size and equally charge/mass ratio but it closely similar with silicon the second member of carbon family and show diagonal relationship.

Several important similarities among boron and silicon are given lower:

(1) Both are non-metals and their melting point; boiling point are high; equal densities (B = 2.35gml^{-1}, S = 2.34 g/ml)

They are bad conductor of electricity and their atomic volumes low, and both are used as semiconductors.

(2) Both are only from covalent compounds.

(3) Both boron and silicon show allotropy property and exist in both crystalline and amorphous form.

(4) Both have nearly equal electronegativity values (B = 2.0, Si = 1.8).

(5) They are form many volatile hydrides and these hydrides on exposure to air they spontaneously catch fire, and they are easily hydrolysed.

(6) Their chlorides are easily hydrolysed by water. Both are equally liquid, fume in moist air.

$$BCl_3 + 3H_2O \rightarrow B(OH)_3 + 3HCl$$

$$SiCl_4 + H_2O \longrightarrow Si(OH)_4 + 4HCl$$

(7) They form weak acids like H_3BO_3 and H_2SiO_3.

(8) They form binary compounds when we treated with metals and in result, they formed borides and silicides. These borides and silicides give mixture of boranes and silanes when we treated with H_3PO_4.

$$3Mg+2B \longrightarrow \underset{\text{(Magnesium boride)}}{Mg_3B_2}\ ; \qquad Mg_3B_2+H_3PO_4 \longrightarrow \text{Mixture of boranes}$$

$$2Mg + Si \longrightarrow \underset{\text{(Magnesium silicide)}}{Mg_2Si}\ ; \quad Mg_2Si + H_3PO_4 \longrightarrow \text{Mixture of silanes}$$

(9) Silicon and boron form carbides and these carbides are very tough and used as abrasive.

(10) Their oxides are acidic in nature, and they can reduced limited amount of Mg but in case when Mg in available into excess amount it formed silicide and boride.

$$B_2O_3+3Mg \longrightarrow 3MgO+2B$$

$$SiO_2+2Mg \rightarrow 2MgO+Si$$

(11) Silicon and boron and their oxides are easily soluble in bases.

$$2B + 6NaOH \longrightarrow \underset{\text{(borate)}}{2Na_3BO_3} + 3H_2 \uparrow$$

$$Si + 2NaOH + H_2O \longrightarrow \underset{\text{(silicate)}}{Na_2SiO_3} + 2H_2^- \uparrow$$

$$B_2O_3 + 6NaOH \rightarrow 2Na_3BO_3 + 3H_2O$$

$$SiO_2 + 2NaOH \rightarrow Na_2SiO_3 + H_2O$$

In silicates and borates have tetrahedral structure units BO_4^{n-} and SiO_4^{n-} respectively. In case boron switches silicon is the three-dimensional lattice it is known as borosilicate's, while boron have planar BO_3 units.

(12) Their acids form volatile esters on heating up with alcohol in presence of concentrated H_2SO_4.

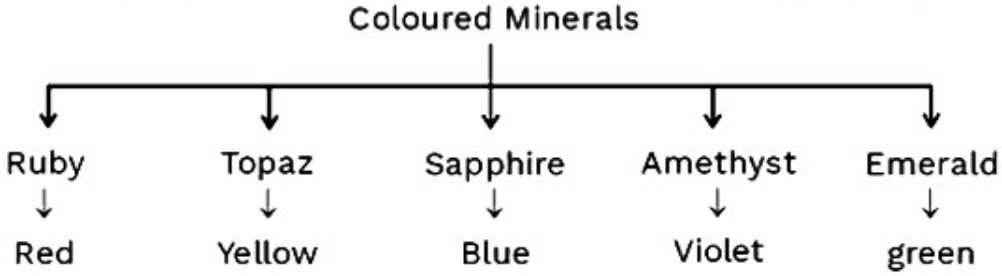
$Si(OH)_4 + 4ROH \rightarrow Si(OR)_4 + 4H_2O$

$B(OH)_3 + 3ROH \rightarrow B(OR)_3 + 3H_2O$

Compounds of Aluminium

(i) Aluminium oxide or Alumina (Al_2O_3) :

In nature as colourless corundum and some coloured minerals.



These minerals are utilized as precious stones or precious gems.

Q12 **Aluminium is stable in air and water. Why?**

Sol. The thin protective film of oxide Al_2O_3 is formed which protects the metal from further attack of air and water and thus it is stable in air and water.

Q13 **Pure alumina can not be electrolysed. Why?**

Sol. Pure alumina Al_2O_3 is covalent in nature, so it can't be electrolysed.

(ii) Aluminium chloride (Al_2Cl_6):

When we are passing dry chlorine over aluminium powder it formed aluminium chloride.

$$Al_2O_3 + 3C + 3Cl_2 \rightarrow \underset{\text{(anhydrous)}}{2AlCl_3} + 3CO(g)$$

It exists in inert organic solvents and in vapour state as a dimer form under vacuum at 100°C it sublimes. When $AlCl_3$ dissolved in water (H_2O), its dimeric structure disappears.

It absorbed moisture while exposed to air and it is hygroscopic in nature.

(iii) Thermite:

If we mix aluminium power and Fe_2O_3 in the ratio of 1 : 3 we get thermite and we use it for welding of iron. Reaction between Al and Fe_2O_3 is exothermic in nature.

$$Al + Fe_2O_3 \rightarrow Al_2O_3 + Fe + Heat$$

(iv) Aluminium sulphate [$Al_2(SO_4)_3$]:

(a) It is use for making fireproof clothes

(b) For formation of alums

Example: Potash Alum $K_2SO_4.Al_2(SO_4)_3.\ 24H_2O$.

ALUM:

(a) In general, the term alums is given to double sulphates of the type:

$$M_2SO_4.M'_2(SO_4)_3.24H_2O$$

$$MM'(SO_4)_2.12H_2O$$

Where M = monovalent cation: Li^+, Na^+, K^+, Rb^+, Cs^+

M' = trivalent cation: Al^{+3}, Cr^{+3}, Fe^{+3}, Mn^{+3}, Co^{+3}

In crystal of alum total 12 molecules of water held by structure 6 water molecules held by monovalent ion and 6 water molecules held by trivalent cation.

(b) Cationic hydrolysis of trivalent cation its aqueous solution of alums is acidic in nature. All alums are isomorphous.

(c) Feather alum or 'Hair-salt' $Al_2SO_4.18H_2O$ is a native form of aluminium sulphate.

(d) Uses:

(i) Tanning of leather
(ii) Dyeing
(iii) Calico printing for sizing paper
(iv) Purification of water
(v) As a styptic to stop bleeding

$K_2SO_4.Al_2(SO_4)_3.24H_2O$ = Potash alum

$(NH_4)_2SO_4.Al_2(SO_4)_3.24H_2O$ = ammonium alum

$K_2SO_4.Cr_2(SO_4)_3.24H_2O$ = chrome alum

$(NH_4)_2SO_4.Fe_2(SO_4)_3.24H_2O$ = ferric alum

If trivalent M' is Al^{+3}, name will be given according to monovalent atom but if Al is not there then name will be given according to trivalent metal.

(e) **Pseudo alums:** Double sulphates of divalent ions and trivalent ions with 24 water molecules in their crystals are known as Pseudo alums. General formula is $MSO_4.M'_2(SO_4)_3.24H_2O$.

M = Bivalent metal; **M'** = Trivalent metal

(f) Pseudo alums are not isomorphous with alums.

Q14 **Which of the following is pseudo alum?**

(A) $K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$ **(B)** $(NH_4)_2SO_4 \cdot Mn(SO_4)_3 \cdot 24H_2O$

(C) $Na_2SO_4.Fe_2(SO_4)_3.24H_2O$ **(D) None of these**

Sol. **(B)**

In $(NH_4)_2SO_4 \cdot Mn(SO_4)_3 \cdot 24H_2O$, here oxidation number of Mn is +6.

Heating effect of alum:

$$K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O \xrightarrow[-24H_2O]{\Delta} K_2SO_4 \cdot Al_2(SO_4)_3 \xrightarrow[\text{temperature}]{\text{high}} \underbrace{K_2SO_4 \cdot Al_2O_3 + SO_3 \uparrow}_{\text{Amorphous powder}}$$

Q15 **Aluminium vessels should not be cleaned by washing soda. Why?**

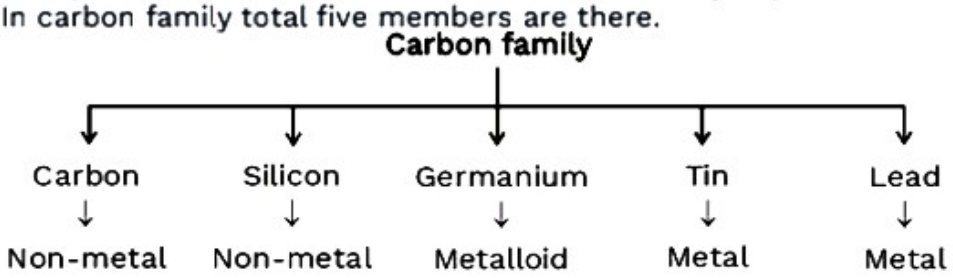
Sol. Na_2CO_3 undergo hydrolysis to give NaOH Al dissolves in caustic alkalies.

$$2Al + 2NaOH + 2H_2O \rightarrow \underset{\text{sod. meta aluminate (solutble)}}{2NaAlO_2} + 3H_2$$

Introduction to Group 14 Elements : Carbon Family

Carbon Family

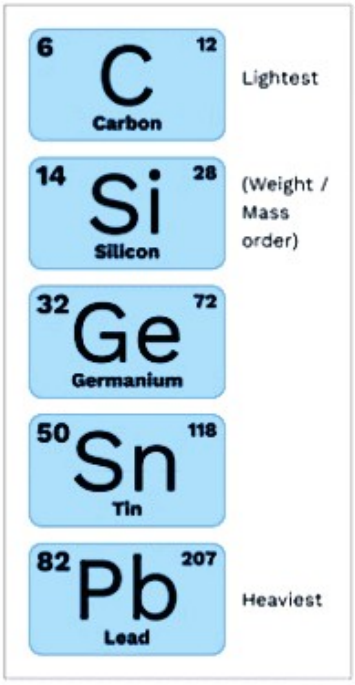
In periodic table group 14 is known as carbon family and in carbon family carbon is first element or first member of group 14.
In carbon family total five members are there.



Group 14 Elements Electronic Configuration:

Their general electronic configuration is "ns^2np^2"

Element	Electronic Configuration
$_{6}C$	$[He]2s^2 2p^2$
$_{14}Si$	$[Ne]3s^2 3p^2$
$_{32}Ge$	$[Ar]3d^{10} 4s^2 4p^2$
$_{50}Sn$	$[Kr]4d^{10} 5s^2 5p^2$
$_{82}Pb$	$[Xe]4f^{14} 5d^{10} 6s^2 6p^2$



RANGE OF SIZE

Group 14 Elements Physical Properties

(1) Non-metallic nature:

When we go along the group in periodic table non-metallic nature reduces..

C, Si	Ge	Sn	Pb
Non-metals	Metalloid	Soft Metal	Soft Metal

(2) Abundance:

In the earth curst carbon and silicon are most abundant element Tin and lead found in small quantities whereas germanium found only in traces.

In group-14 carbon is only one which is exist in free and in combined state

⇒ In free state – Graphite and Dimond

⇒ In combined state – CO_2, CO, Carbonates, wood, Coal etc

Element	Source
Germanium	Traces of coal
Tin	Tin stone (SnO_2) or cassiterite
Lead	Galena (PbS) Anglesite ($PbSO_4$) Cerussite($PbCO_3$)

Note: Germanium is used in conductors and transistors.

(3) Density:

As going down the group density increase due to molar mass increases.

Element	C	Si	Ge	Sn	Pb
Density (g/ml)	3.51 (for diamond) 2.22 (for graphite)	2.34	5.32	7.26	11.34

(4) Melting point and boiling points:

When we go down in the group their melting point and boiling point decreases.

Element	C	Si	Ge	Sn	Pb
Melting point (K)	4373	1693	1218	505	600
Boiling point (K)	–	3550	3123	2896	2024

(5) Atomic radii and atomic volume:

As going down the group atomic radius increases due to number of shell increases.

Element	C	Si	Ge	Sn	Pb
Atomic radius (pm)	0.77	111	122	141	144

(6) Electronegativity:

As going down the group electronegativity decreases due to metallic characters increase.

Element	C	Si	Ge	Sn	Pb
Electronegativity on Pauling scale	2.5	1.8	1.8	1.7	1.6

In case of Si and Ge E.N. become constant due to d10 stable electronic configuration.

(7) Ionisation energy

The ionisation energy reduces regularly down the group.

Note:

Pb show higher value of ionisation energy than Sn this is happens because poor shielding of inner f-orbitals because of that effective nuclear charge experienced by outer shell electrons becomes more in Pb.

Ionization energy: (kJ/mol^{-1})	C	Si	Ge	Sn	Pb
IE_1	1086	786	761	708	715
IE_2	2352	1577	1537	1411	1450

(8) Oxidation state:

(i) Due to 4 valence electrons they can form M^{4+} , therefore they are showing ionic as well as covalent characters by sharing 4 electrons.

(ii) Formation of M^{4+} require large amount of energy therefore, these elements generally do not form M^{4+} or M^{4-} ions, but they generally form compounds with covalence of four.

(iii) Due to inert pair effect Ge, Sn and Pb also show +2 oxidation state.

(iv) Sn^{2+} and Pb^{2+} both are exhibit ionic nature.

(v) Due to inert pair effect when we go down in the group the tendency to show +2 oxidation state increases.

(9) Catenation:

The property of forming bonds with atoms of the same clement or tendency to self linking is called catenation. Carbon shows maximum catenation. On moving down the group, catenation tendency decreases. This is because the strength of C – C bond is very high. Carbon atoms similarly form double or triple bonds including pπ-pπ multiple bonds within itself.

> C = C < ; – C ≡ C –

Strength of M – M (where M = Si, Ge, Sn, Pb) bond decreases down the group.

Bond	Bond Energy (kJ mol^{-1})
C – C	348
Si – Si	297
Ge – Ge	260
Sn – Sn	240

(10) Allotropy:

Those elementary forms which have Different physical forms are known as allorops.

Note:

Just like allotropy, If an element or compound exists in two or more forms, it is also known as polymorphism e.g. zinc blende and wurtzite are polymorphs of ZnS. In carbon family, except lead, all elements show allotropy.

Allotropes of carbon

Carbon has two types of Allotropic forms :

(1) Crystalline (2) Amorphous

(1) Crystalline: Diamond, Graphite and Fullerene

(i) Diamond:

1. Each carbon is linked to another carbon atom in closed packing in structure of Diamond.
2. Diamond has 3D network solid of covalent bonds so that it is very hard and dense. Every carbon is SP3 hybridise and due to this it is tetrahedral arrangement.

3. It is employed in cutting tool for of glass due to sharp edges.
4. Diamond is a non-conductor of electricity due to no free electron available..

(ii) Graphite:

1. In graphite carbon atoms are **sp^2 hybridized** and due to this, graphite exist as hexagonal layer.
2. Each one carbon is linked with 3 carbons forming a **two dimensional sheet like structure.**
3. Each layers are connected by weak vend wall force therefore each layers are sliding on each other.
4. Graphite is a conductor of electricity due to presence of unpaired electron.
5. C-C bond length in Graphite is shorter (1.42Å) than that of Diamond (1.54 Å).
6. Graphite used in manufacture of Crucible due to high melting and boiling point.
7. Graphite reduction tendency when heated like alkaline forms metallic acid.

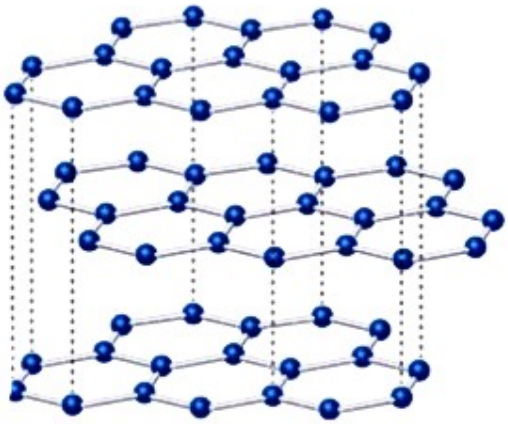
(iii) Buck Minster Fullerene:

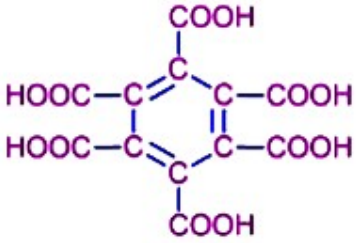
1. It has the formula C_{60} and is made from interlocking hexagonal and pentagonal rings of carbon atoms.
2. Such molecules are now thought to exist even in chimney root or candle smoke.
3. The structure of C_{60} is similar to the surface of a football which has also set of interlocking hexagons and pentagons.
4. Another molecule C_{70} has been recently discovered.
5. These similar large carbon molecules are sometimes referred as "bucky balls".

Diamond

In the diamond structure each carbon atom forms four covalent bonds with four other carbon atoms to form a 3D tetrahedral structure

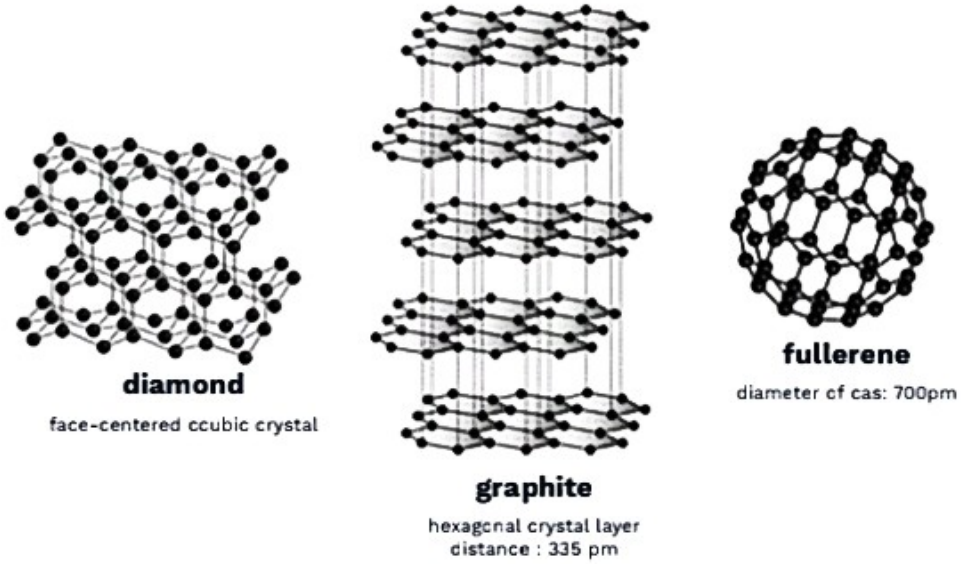






Benzene hexa carboxylic acid





2. Amorphous Allotropic Forms of Carbon

(A) Lamp Black:

(i) It is obtained by incomplete combustion of compounds which contains higher % of carbon: Benzene, turpentine, acetylene etc. These all on combustion form black carbon called lamp black.

(ii) Black blue ink, printing ink, black paints, varnishes are made from lamp black.

(B) Coke:

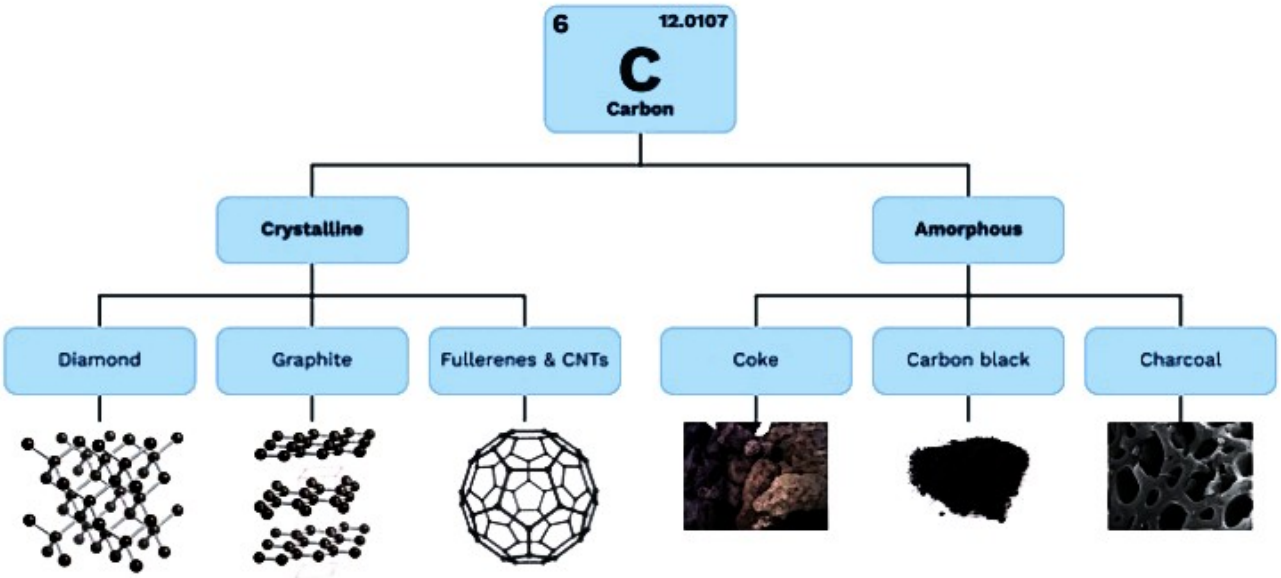
(i) It obtained by destructive distillation of wood.

(ii) Coke is usually employed as weak reducing agent in comparison to CO.

(C) Wood Charcoal:

(i) It obtained by incomplete combustion of wood.

(ii) It is used to decolorize organic compound.



Q16 **Carbon dioxide does not support combustion but a burning magnesium ribbon continues to burn in it.**

Sol. Oxygen support for combustion not CO_2.

Fire obtained by Mg is highly exothermic reaction

$2Mg + CO_2 \rightarrow 2MgO + C$

Q17 **C—C bond length in graphite is shorter than C—C bond of diamond.**

Sol.

Diamond	Graphite
(C–C)	(C–C)
$sp^3 - sp^3$	sp^2–sp^2 Hybridized.
B.L. = 1.54Å	B.L. = 1.42Å

In graphite more (S) character in hybridized atom., smaller size of Hybridized orbital more is the overlapping. Shorter is the bond length

Allotropic forms of Tin

(i) There are three allotropic forms of exist of Tin.

$$\underset{\text{(Grey)}}{\alpha - \text{tin}} \xleftarrow{18^\circ C} \underset{\text{(White)}}{\beta - Sn} \xrightarrow{160^\circ C} \underset{\text{(Rhombic)}}{\gamma - Sn}$$

(ii) The formation of grey tin is carried out by increase in volume and the latter, being very brittle, easily crumbles down to powder. This process is called tin disease, tin pest or tin plague.

(iii) When tin sheets are folded. they give a peculiar sound which is called as **Tin cry.**

(iv) White tin used in containers of oil. White tin is more stable and having maximum density.

Compounds of carbon family

Hydrides

(i) All the members of carbon family combine with hydrogen directly to form the covalent hydrides, MH_4 **(M = C, Si, Ge, Sn or Pb).** Carbon form large number of hydrides like alkane and alkenes.

(ii) The hydrides of silicon are known as silanes and the general formula Si_nH_{2n+2}. Silanes with 'n' up to 8 are commonly known. Germane are the hydrides of germanium while stannanes are the hydrides of tin. Only lead formed an unstable hydride, known as the plumbane.
Only three hydrides of germanium are stable, i.e., GeH_4, Ge_2H_6, Ge_3H_8 and only two hydrides of tin i.e., SnH_4 and Sn_2H_6 are well known.

(iii) Thermal stability of hydrides decreases down the group.

Oxides

(i) Carbon forms various oxides commonly carbon monoxide and carbon dioxide. CO_2 is a non-polar in nature and linear molecule due to max. tendency of Carbon to form $p\pi - p\pi$ multiple bond with oxygen.

(ii) Si has silicon oxide SiO_2.

(iii) Pb forms various of oxides. PbO prepared by lead nitrate..
$2Pb(NO_3)_2 \xrightarrow{Heat} 2PbO + 4NO_2 + O_2$
Pb_3O_4 is a mixed oxide of PbO_2 and 2PbO. Pb_2O_3 is known as lead sesquioxide.

(iv) GeO_2, SnO_2 etc. are also found as network solid.

(v) CO_2 and SiO_2 is acidic, GeO_2 is weakly acidic while SnO_2 and PbO_2 are amphoteric in nature.

(a) Carbon monoxide CO:

(i) By incomplete combustion of carbon

$2C + O_2 \rightarrow 2CO$

(ii) Reduction of oxides of heavy metals with carbon.

$ZnO + C \rightarrow Zn + CO$

$Fe_2O_3 + 3C \rightarrow 2Fe + 3CO$

$PbO + C \xrightarrow{\Delta} Pb + CO$

(iii) Laboratory Method: By dehydration of formic acid with conc. H_2SO_4

$$\underset{\text{Formic acid}}{HCOOH} \xrightarrow[373K]{H_2SO_4} \underset{\text{(pure)}}{CO} + H_2O$$

(iv) By action of conc. H_2SO_4 on potassium ferrocyanide

$$\underset{\text{Potassium ferrocyanide}}{K_4\left[Fe(CN)_6\right]} + 6H_2SO_4 + 6H_2O \xrightarrow{\Delta} 2K_2SO_4 + 3(NH_4)_2SO_4 + 6CO$$

Carbon monoxide
CO

C Carbon

O Oxygen



Q18 **Which of the following combinations is obtained on heating potassium ferrocyanide with concentrated H_2SO_4?**

(A) CO_2 **(B) CO** **(C) C_2H_2** **(D) $(CN)_2$**

Sol. **(B)**

$$K_4\left[Fe(CN)_6\right] + 6H_2SO_4\,(conc) + 6H_2O \xrightarrow{\Delta} 2K_2SO_4 + FeSO_4 + 3(NH_4)_2SO_4 + 6CO$$

Physical Properties:

(i) Carbon monoxide is a neutral oxide.
(ii) It is colourless gas.
(iii) It is slightly soluble in water.
(vi) It gives blue colour to flame.

Note:

Poisonous nature due to its ability to form a stable complex with the haemoglobin present in the red blood cells. CO has 300 times more tendency than O_2 to bind with haemoglobin forming carboxyl haemoglobin.

Haemoglobin + CO → Carboxy haemoglobin

Due to formation of stable complex with CO, Haemoglobin looses its O_2 carrying capacity which causes death.



Chemical properties:

(i) Formation of metal carbonyls

$$\underset{\text{Nickel carbonyl}}{Ni + 4CO} \xrightarrow{330-350K} Ni(CO)_4$$

$$\underset{\text{Iron carbonyl}}{Fe + 5CO} \xrightarrow[100\,atm]{473K} Fe(CO)_5$$

(ii) CO is readily absorbed by a soln. of CuCl in concentrated HCl or NH_3 due to the formation of solution complexes.

$$CuCl + NH_3 + CO \rightarrow \underset{\text{(soluble complex)}}{\left[Cu(CO)NH_3\right]^+ Cl^-}$$

$$CuCl + HCl + CO \rightarrow \underset{\text{(soluble complex)}}{H^+\left[Cu(CO)Cl_2\right]^-}$$

(iii) Reaction with Cl_2

$$CO + Cl_2 \xrightarrow{\text{sunlight}} COCl_2$$

(iv) Reaction with NaOH

$$CO + NaOH \xrightarrow[\text{Low T}]{\text{Low P}} HCOONa$$

Estimation of Carbon Monoxide :

(i) It reduces iodine pentoxide to iodine and this iodine can be titrated with standard sodium thiosulphate solution. CO can be estimated by this method.

$$5CO + I_2O_5 \rightarrow I_2 + 5CO_2$$

(ii) CO does not support combustion but itself burn in air with a blue flame producing carbon suboxide.

Tests of carbon monoxide:

(i) It burns with blue flame.

(ii) Filter paper soaked with PCl_2 is turned pink, green or black. $PdCl_2$ is reduced to the metal.

(iii) It reduces iodine penta oxide into free iodine, which gives colour with CCl_2, CS_2 or $CHCl_2$.

Syn gas $\quad C + H_2O \xrightarrow{473-1273K} \underbrace{CO + H_2}_{\text{Water gas/ Syn gas/synthesis gas}}$

Producer gas $\quad \underbrace{C + O_2 + N_2}_{\text{air}} \longrightarrow \underbrace{CO + N_2}_{\text{producer gas}}$

Q19 **Which is better fuel?**
(A) Water gas > Producer gas **(B) Water gas < Producer gas**

Sol. **(A)**

Water gas is better fuel because in producer gas, N_2 is inert which is not helpful.

(b) Carbon dioxide CO_2:

Preparation:

(i) By burning carbon, fossil fuels and other organic compounds in air.

$$C + O_2 \rightarrow CO_2$$

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$$

(ii) Laboratory method: By action of HCl on carbonates

$$CaCO_3 + 2HCl \xrightarrow{\Delta} CaCl_2 + CO_2 + H_2O$$

$$NaHCO_3 + HCl \rightarrow NaCl + H_2O + CO_2$$

Carbon dioxide
Co_2

C Carbon

O Oxygen



Physical Properties:

(i) Colourless & odourless gas (about 1.5 times heavier than air).
(ii) It is not poisonous but does not support life.

Chemical properties:

(i) Noncombustible nature (certain active metals Na, K, Mg burn in it)

$$2Mg + CO_2 \rightarrow 2MgO + C$$

$$Zn + CO_2 \rightarrow ZnO + CO$$

(ii) Acidic Nature: It dissolves in water to some extent to form carbonic acid.

$$CO_2 + H_2O \rightarrow \underset{\substack{\text{Carbonic acid}\\(\text{weak dibasic acid})}}{H_2CO_3} \rightleftharpoons H^+ + HCO_3^- \rightleftharpoons 2H^+ + CO_3^{-2}$$

(iii) It reacts with metallic oxides

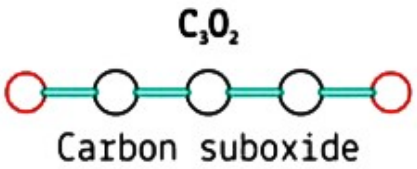
$$CaO + CO_2 \rightarrow CaCO_3$$

$$Na_2O + CO_2 \rightarrow Na_2CO_3$$

(c) Carbon suboxide C_3O_2:

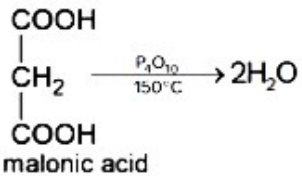
Structure:

$O = C = C = C = O$ (Linear)



Preparation:

It is made by dehydrating malonic acid with P_4O_{10}.



Properties:

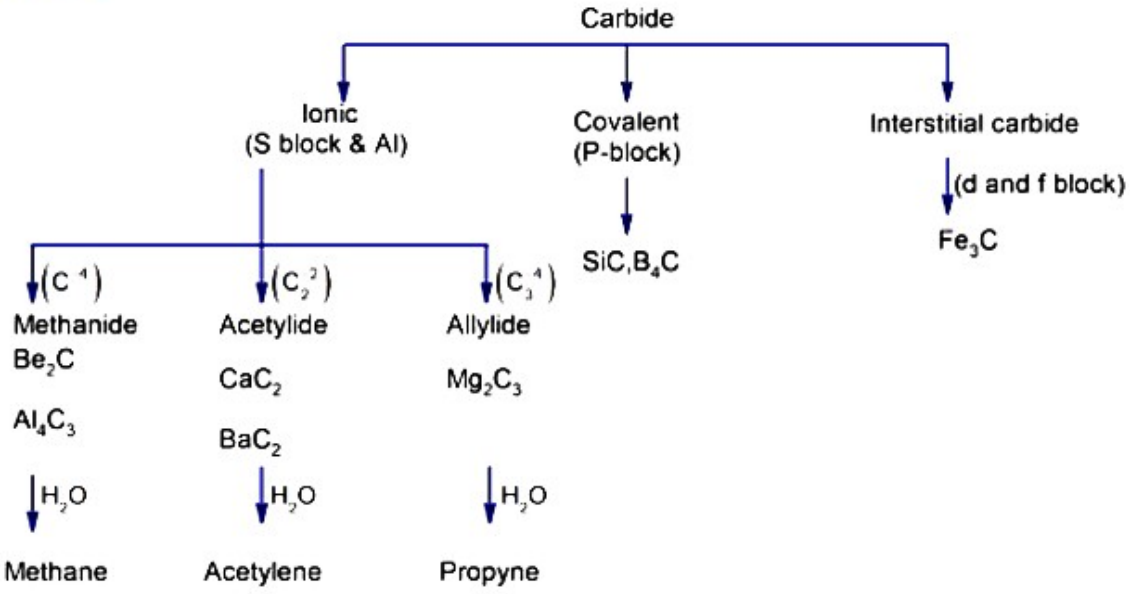
(i) Colourless gas with b.p. 6°C.

(ii) Very readily polymerises to give coloured salts.

(iii) It decomposes at 200°C.

$$C_3O_2 \xrightarrow{200°C} CO_2 + 2C$$

(3) Carbide



e.g. $Be_2C + H - OH \rightarrow Be(OH)_2 + CH_4$

$Al_4C_3 + H - OH \rightarrow Al(OH)_3 + CH_4$

$CaC_2 + H - OH \rightarrow Ca(OH)_2 + CH \equiv CH$

$$BaC_2 + H-OH \rightarrow Ba(OH)_2 + CH \equiv CH$$

$$Mg_2C_3 + H-OH \rightarrow Mg(OH)_2 + CH_3 - C \equiv CH$$

Types of Carbide:

(i) Ionic or salt-like:

Classification on the basis of no. of carbon atoms present in hydrocation found on their hydrolysis.
- (a) C_1 unit
- (b) C_2 unit
- (c) C_3 unit

(a) C_1 unit: Al_4C_3, B_2C

$$Be_2C + H_2O \rightarrow Be(OH)_2 + CH_4$$

$$Al_4C_3 + 12H_2O \rightarrow 4Al(OH)_3 + 3CH_4$$

(b) C_2 unit: CaC_2, BaC_2

$$CaC_2 + 12H_2O \rightarrow Ca(OH)_2 + CH \equiv CH$$

(c) C_3 unit: Mg_2C_3

$$Mg_2C_3 + H_2O \rightarrow 3Mg(OH)_2 + CH_3 - C \equiv CH$$: Propyne

(ii) Covalent carbide: **SiC & B_4C**

(iii) Interstitial carbide: Transition element or inner transitional elements forms this kind of carbide.
Interstitial carbide formation doesn't affect the metallic lusture and electrical conductivity.

4. Halides

(i) They form tetra halides MX_4 except $PbBr_4$ & PbI_4.

(ii) The nonexistence of $PbBr_4$ & PbI_4 is due to that Pb^{+4} is a strong oxidizing agent while Br^- & I^- are highly reducing agent.

(iii) All MX_4 are covalent except SnF_4. It is Ionic.

(iv) The tetra halides of carbon cannot undergo hydrolysis due to non-availability of vacant 'd' orbitals $\underset{\text{phosgene}}{CCl_4} + H_2O \rightarrow COCl_2 + 2HCl$

(v) The tetra halides like SiF_4 can form SiF_6^{2-}. In this ion, silicon undergoes sp^3d^2 hybridization where carbon cannot form this type of ion.

(vi) Dihalides of these elements are more ionic than their corresponding tetra halides.

(vii) Thermal stability of tetra halides:

$CX_4 > SiX_4 > GeX_4 > SnX_4 > PbX_4$

(viii) Stability of dihalides:

$CX_2 < SiX_2 < GeX_2 < SnX_2 < PbX_2$

Q20 **Which one is more co-valent?**

(A) $MX_2 > MX_4$ **(B) $MX_2 < MX_4$**

Sol. **(B)**

Higher the charge, higher will be polarisation, so higher the covalent character.

$MX_2 < MX_4$

Q21 **Which is more volatile?**

(A) $MX_2 < MX_4$ **(B) $MX_2 > MX_4$**

Sol. **(B)**

Covalent compounds are more volatile.

Q22 **Which of the following halide does not exist?**

(A) SnF_4 **(B) PbF_4** **(C) PbI_4** **(D) All**

Sol. **(C)**

Lead tetraiodide, PbI_4 does not exist because the iodide ion cannot stabilize the +4 oxidation state of lead.

5. Oxyacids:

(i) Oxyacids are formed by C & Si.

(ii) Main inorganic oxyacid is H_2CO_3.

(iii) H_2SiO_3 is meta silicic acid.

Q23 **Which element-element has the highest bond dissociation energy?**
(A) C – C **(B) Si – Si** **(C) Ge – Ge** **(D) Sn – Sn**

Sol. **(A)**

C–C bond dissociation energy is 355 kJ mol^{-1} which is highest amongst the carbon family members.

Q24 **In graphite, electrons are**
(A) Localized on every third C-atom
(B) Present in anti-bonding orbital
(C) Localized on each C-atom
(D) Spread out between the structure.

Sol. **(D)**

In graphite, each carbon is sp^2 hybridized and forms four covalent bonds with other C-atoms overlapping sidewise to give π–electron cloud which is delocalized and thus the electrons are spread out between the structure.

Q25 **Consider following statements:**
I. In diamond, each one carbon atom is linked tetrahedrally to four other carbon atoms by sp^3 bonds.
II. Graphite has planar hexagonal layers of carbon atoms held together by weak Vander Waals forces
III. Silicon exists only in diamond structure due to its tendency to form pπ—pπ bonds to itself.
(A) Only I and II are correct **(B) Only I is correct**
(C) Only II and III are correct **(D) All are correct statements**

Sol. **(D)**

Q. **Select correct statement:**
(A) Oxides of carbon family (MO_2) are all network solids with octahedral coordination.
(B) Silicon dioxide (silica) is a network solid with tetrahedral coordination and is a giant molecule.
(C) GeO_2, SnO_2 and PbO_2 are all network solids with octahedral coordination
(D) None appears correct.

Sol. **(BC)**

Anomalous behaviour of Carbon

Carbon shows different properties this due to-

- Its small size and higher electronegativity.
- Absence of d-orbitals
- Maximum covalency i.e. 4
- Catenation property

C-C > Si-Si > Ge-Ge > Sn-Sn

Carbon shows maximum catenation property due to C-C bond strength is very high and moving down to the group bond strength decreases.

Bond	Bond energy (K\kj/mole)
C-C	348
Si-Si	297
Ge-Ge	260
Sn-Sn	240

Some of the properties in which it varies from other representatives are:

(1) Higher melting and boiling point with respect to other element.
(2) Hardest element i.e. diamond form by carbon.
(3) Maximum covalence with respect to other element.
(4) Due to no availability of d-orbitals it form Pπ–Pπ multiple bonds while other have tendency to form Pπ–dπ multiple bonds.
(5) Oxide of carbon found in gaseous state while other found in solid state.
(6) Carbon is unaffected affected from alkalies while other reacts with basees

For example, silicon form silicates.

$$Si + 2NaOH + 1/2\,O_2 \rightarrow \underset{\text{Sodium silicate}}{Na_2SiO_3} + H_2$$

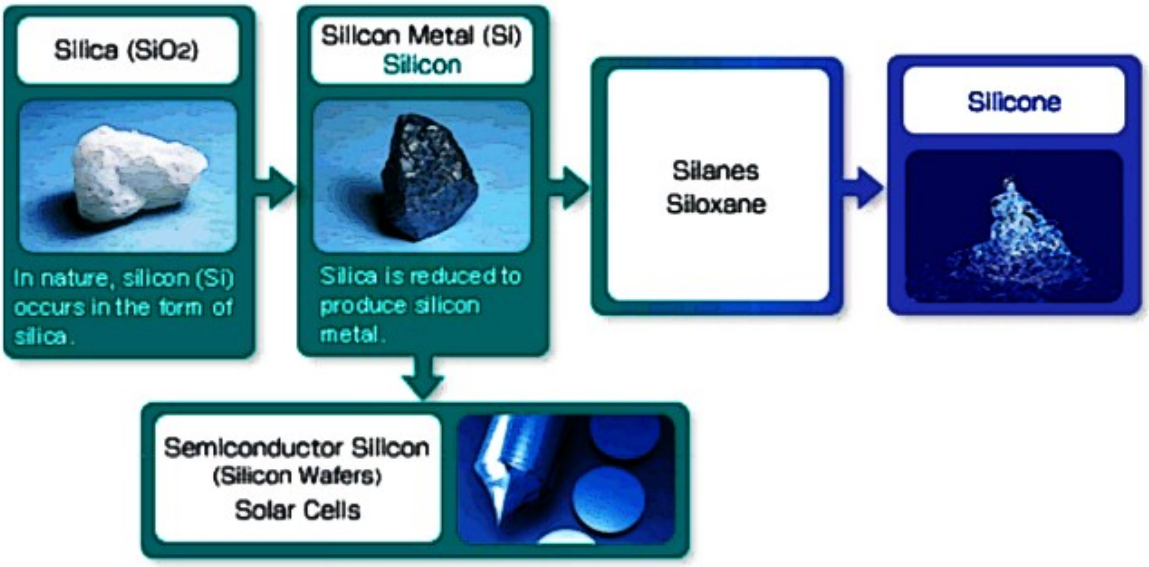
SILICON (Si)

Occurrence:

The abundance of Silicon(Si) 27.2% which is 2nd abundant element in earth crust after oxygen (45.3%). It does not found in free but exist in combined state like silica and silicates.

Some of minerals of silicon as-

(i) Feldspar : $K_2O.Al_2O_3.6SiO_2$
(ii) Kaolinite : $Al_2O_3.2SiO_2.2H_2O$
(iii) Asbestos : $CaO.\ 3MgO.4SiO_2$



Method of Preparation:

(i) By reduction of silica.

$$SiO_2 + 2C \longrightarrow Si + 2CO$$

(ii) By reduction of reduction of silicon tetrachloride and silicon chloroform

$$SiCl_4 + 2H_2 \longrightarrow Si + 4HCl$$

$$SiHCl_3 + H_2 \longrightarrow Si + 3HCl$$

Chemical Properties:

(i) Silicon has three isotopes ${}_{14}Si^{28}$, ${}_{14}Si^{29}$, ${}_{14}Si^{30}$ and most commonly used is ${}_{14}Si^{28}$.

Note : Silicon do not react at room temperature towards most of the elements except fluorine.

(ii) Action of air

React with oxygen at 1173 K and reacts with nitrogen at 1673K and form silicon oxide and silicon nitride respectively

$$Si + O_2 \rightarrow SiO_2$$

$$3Si + 2N_2 \rightarrow Si_3N_4$$

(iii) Action of steam

$$Si + 2H_2O \xrightarrow{\text{Redness}} SiO_2 + 2H_2$$

(iv) Reaction with halogens

Reaction occurs at room temperature in case of fluorine, except fluorine other halogens required higher temperature.

$Si + 2F_2 \rightarrow S_2F_4$

(v) Reaction with carbon

$Si + C \xrightarrow{2500K} SiC$

In this reaction carborundum is form which is very hard and inert and used as refractory material.

(vi) Reaction with base

$Si + 4NaOH \xrightarrow{Heat} Na_4SiO_4 + 2H_2 \uparrow$

(viii) Reaction with carbonates

$Si + Na_2CO_3 \longrightarrow Na_2SiO_3 + C$

Uses of silicon:

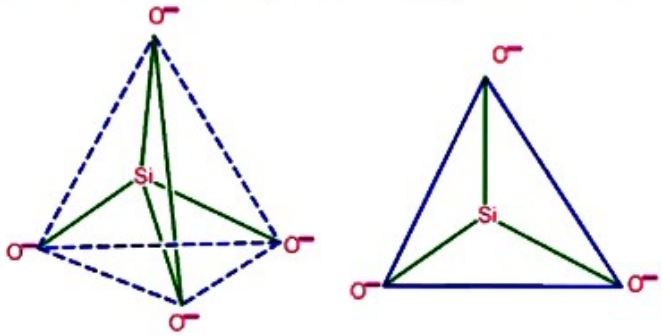
(i) Highly pure silicon used in the semi-conductors like transistors.

(ii) It is used as acid resistance in formation of alloys.

Compounds of Silicon

(i) Silicates

Clay and silica mainly form the outermost covering of earth .These are silicate minerals. These are considered as metal products of silicic acid (H_4SiO_4). Each silicate unit contains SiO_4^{4-} anion. In SiO_4^{4-} anion, Si is sp^3 hybridised and bonded to four negatively charged oxygen atoms with covalent bonds. The shape of the SiO_4^{4-} anion is tetrahedral shape.



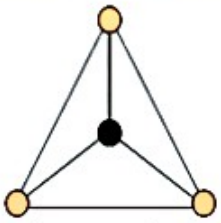
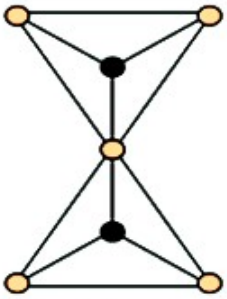
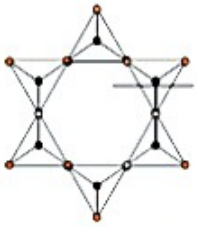
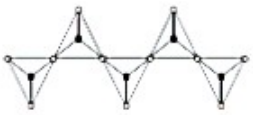
Preparation:

1. By alkali metal

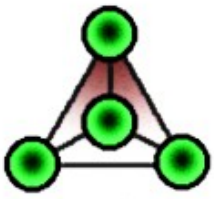
$$\underset{\text{sand}}{Na_2CO_3 + SiO_2} \xrightarrow{\text{Fusion}} \underset{\text{sodium silicate}}{Na_2SiO_3} + CO_2(g)$$

This reaction occurs at high temperature.

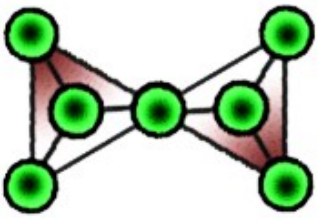
Different types of Silicates

Zero	Zero	 Discrete SiO_4^{4-} anion	Si = + 4 O = − 8 Net = −4 SiO_4^{4-}	Ortho-silicates Mg_2SiO_4
1	1	 Island structure	Si = + 8 O = − 14 Net = −6 $(Si_2O_7)^{6-}$	Pyro- silicates
2	2	Ring anion	Si= +12 O = −18 Net = −6 $(Si_3O_9)^{6-}$	Wollas-tonite $Ca_3Si_3O_9$
2	2	 Ring anion	Si = +24 O = −36 Net = −12 $(Si_6O_{18})^{12-}$	Beryl, $Be_3Al_2Si_6O_{18}$
2	2	 Chain anion	Si = +4 O = −6 Net = −2 $(SiO_3^{2-})_n$	Pyroxenes, e.g., MgCa Si_2O_6 Asbestos

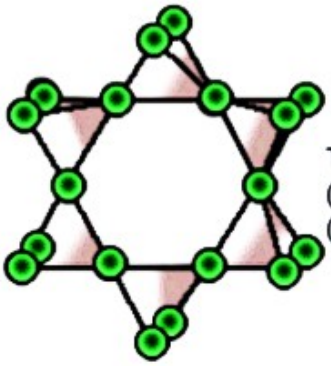
No. of corners of SiO_4 tetrahedral shared	No. of common oxygen atoms	Structure		Net charge and Anion in the silicate structure	Example
		Diagram and Description			
3	3	 Two–dimensional sheet structure		Si = +8 O = −10 Net = −2 $\left(Si_2O_5^{2-}\right)_n$	Clays, talc kaolinite
4	4	Three–dimensional network		Si = +4 O = −4 $\left(SiO_3\right)_n$	Quartz, Tridymite and Cristo-balite



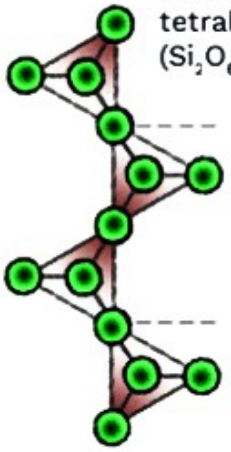
Isolated Tetrahedron
$(SiO_4)^{4-}$



Double Tetrahedra
$(Si_2O_7)^{6-}$

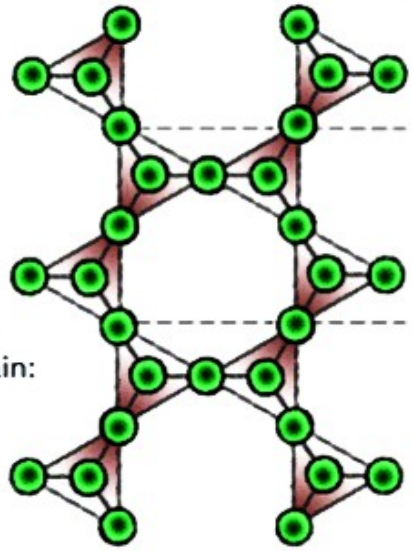


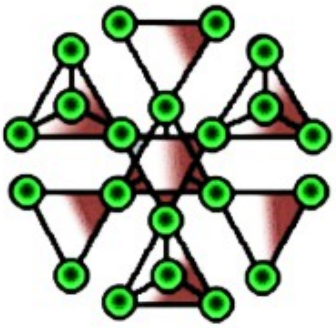
Tetrahedral ring (6-fold):
$(Si_6O_{18})^{12-}$



Infinite, single tetrahedral chain:
$(Si_2O_6)^{4-}$

Infinite, double tetrahedral chain:
$(Si_4O_{11})^{6-}$





Infinite tetrahedral network:
$(SiO_2)^0$

(ii) Silica or silicon dioxide (SiO_2)

It has network or polymer structure. It is not iso structural like CO_2 due to presence of weak π-bond.

Preparation:

Silicon dioxide form by reaction of air with silicon at 1173 K temperature and further it will polymerise and convert in silica.

$Si(s) + O_2(g) \rightarrow SiO_2(s)$

Properties:

- It is Amorphous solid (non crystalline) .
- It is also known as super cooled liquid or quartz.
- It is insoluble in water.
- Polymeric silica has 3-D network solid

Note:

It may be noted that CO_2 is a gas while SiO_2 is hard solid with very high melting point.

Reaction with HF

$$SiO_2 + 4HF \longrightarrow SiF_4 + 2H_2O$$

$$SiF_4 + 2HF \longrightarrow \underset{\text{(Hydrc flcurosilicic acid)}}{H_2SiF_6}$$

HF is dissolve in silica so that it can not stored ion glass bottles of silica..

Uses:

- It is main constituent of cement used in building material.
- It is used in the manufacture of glass and lenses.

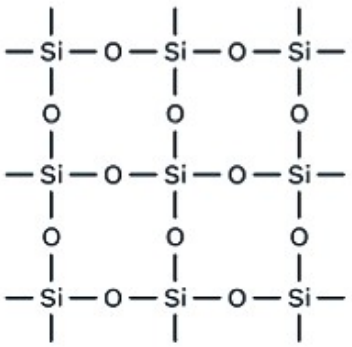
(iii) Silicones

Polymeric compounds containing Si–O–Si bonds with organic group are called silicones. They are also known as organo-silicon compounds just like organometallic compounds. The general formula $(R_2SiO)_n$, where R is CH_3 –group (majority cases) or C_6H_5 – group.

Silicon dioxide

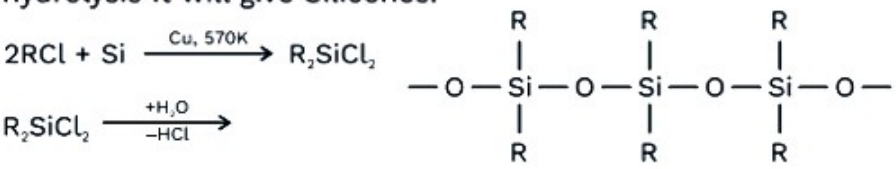
SiO_2



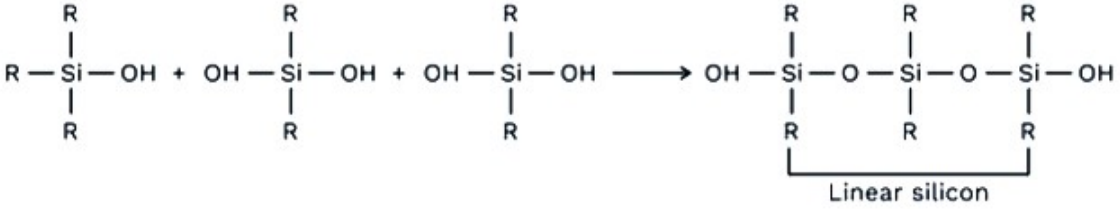


Preparation:

By treating silicon tetra chloride with Grignard which give dialkyl dichlorosilanes (R_2SiCl_2) on hydrolysis it will give Silicones.

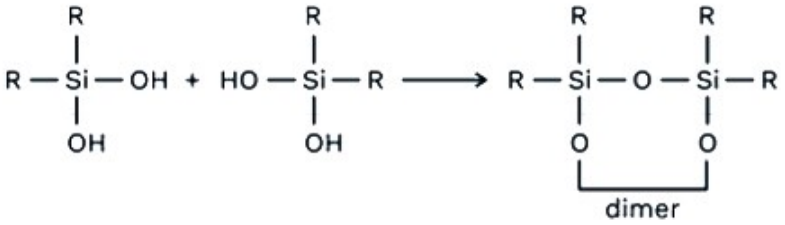


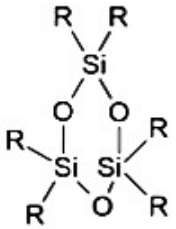
Linear silicon:



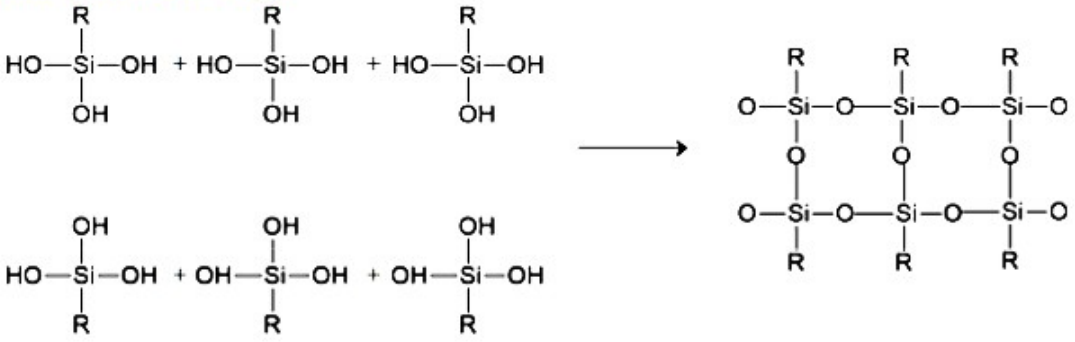
Cyclic silicon:

When in linear silicon, terminal group loses their water then cyclic silicon is formed.





Crosslinked 3D silicane:



Q26 **Which of the following is chain stopping unit in linear Si?**

(A) $R_2SiCl_2 + R_3SiCl$ **(B)** $R_2SiCl_2 + RSiCl_3$

(C) $R_2SiCl_2 + SiCl_4$ **(D)** $R_2SiCl_2 + R_2SiCl_2$

Sol. **(A)**

Q27 **The hydrolysis of CCl_4 is not possible but $SiCl_4$ is easily hydrolysed.**

Sol. Si constant accommodate OH– ion in 3d sub shell whereas 'C' atom doesn't possess d orbital.

- Silicones are inert in nature.
- They are water repellent.
- They are insulator due to inert in nature.

Uses:

(a) They are used in paper and textile industry to manufacturing water-proof papers, wool, fabrics, wood etc.
(b) They are uses in all time weather lubricants due viscosities unaffected with temperature.
(c) They used in various industrial processes as antifoaming agents.
(d) They used in rubber industry as a mould releasing agent.
(e) They used to manufacture of various cosmetics.

(iv) Silanes:

Silicon hydrides are known as silanes and general formula of silanes is Si_nH_{2n+2}.

Examples: (a) SiH_4 (b) Si_2H_6

Preparation:

When vapour of Silicon passed on hot magnesium in presence of dilute sulphuric acid then these silanes will form

Mg (hot) + Si (vap) → Mg_2Si + dil. H_2SO4 → $MgSO_4$ + SiH_4 + Si_2H_6

Properties:

• On heating above silance will decompose, they are less stable then alkanes.

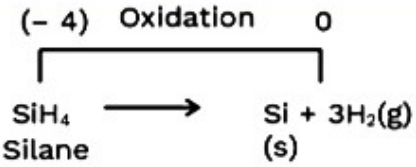
• As molecular mass increases thermal stability decreases.

Uses:

- Used as manufacturing in cosmetics
- To use in sexual health products

8 **Silanes are few in number whereas alkanes are large in number.**

Sol.



Silane are less combustible although, Alkane are highly combustible.

Q29 **Alkanes are more stable than silanes.**

Sol. C – C bond energy is appreciable Higher than Si – Si bond energy and thus alkane are relatively none stable or less than silanet.

(v) Glass:

Glass is **an transparent solid and amorphous** which is achieved by solidification of several silicates and borates of potassium and calcium.

Preparation:

Glass is prepared by sodium carbonate and calcium oxide on treating with silica. The ordinary glass will form.

$$Na_2CO_3 + SiO_2 \longrightarrow Na_2SiO_3 + CO_2\uparrow$$

$$CaO + SiO_2 \longrightarrow CaSiO_3$$

Number of acidic basic and various colouring materials are used as raw material of glass.

Various varieties of glass:

The different types of glass and their constituents are given below:

Type of glass	Constituents	Special use
Soft glass	$Na_2CO_3, CaCO_3, SiO_2$	To making glass tubes, bottles, window panes.

Hard glass	$K_2CO_3, CaCO_3, SiO_2$	To making hard glass apparatus
High refractive index glass (Flint glass)	Lead oxide, K_2CO_3	To making lenses
Pyrex glass	$Na_2CO_3, Al_2O_3, B_2O_3$ or borax, sand	To making laboratory apparatus
Crook's glass	$K_2CO_2, PbCO_3, CeO_2$, sand	To making lenses, and ultraviolet observer apparatus
Jena glass	Zinc & Barium Boro silicates	To making good quality of glass wares.

Coloured glass:

Coloured glasses are made by various metallic oxides of transition metal. Different colure imparting oxides are given below.

Compound added	Colour imparted
Cobalt oxide (CoO)	Blue
Selenium oxide (Se_2O)	Red
Ferric oxide (FeO)	Lemon Yellow
Chromium oxide (Cr_2O_3)	Green
Nickel oxide (NiO)	Black
Manganese dioxide (MnO_2)	Purple

Etching of glass:

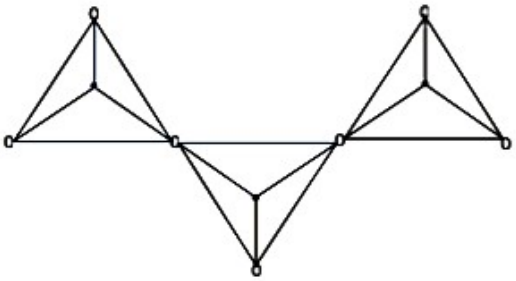
Glass is attacked by hydrofluoric acid. This property is utilized in the etching of glass. The glass to be etched is coated with a thin layer of wax and the design to be made is scratched with a needle. An aqueous solution of HF is applied to the exposed part. After a few time, it is placed in hot water (H_2O) and wax is separated from the surface. The marks are engraved on the exposed parts.

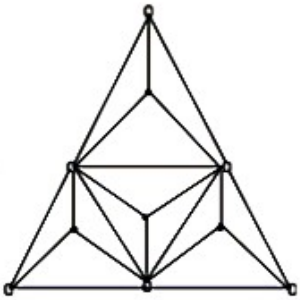
Q30 **Which of the following is/are true about silicones?**
(A) Silicones contain repeated cross-linked silicones.
(B) $RSiCl_3$ gives a complex cross-linked silicones.
(C) $(CH_3)_2SiCl_2$ gives a chain silicone.
(D) When silicones undergo combustion, they form solid white powder as one of the products.

Sol. **(ABCD)**

Combustion of silicones produces solid silica.

Q31 **$Si_3O_9^{6-}$ (having three tetrahedral units) is represented as:**

(A) 

(B) 

(C) Both (A) and (B)

(D) None of these

Sol. **(B)**

(A) 3Si and 10 O (B) 3Si and 9 O

Q32 **The silicate anion in the mineral kinoite is a chain of three SiO_4 tetrahedral unit that share corners with adjacent tetrahedral. The mineral also contains Ca^{2+} ions, Cu^{2+} ions and water molecules in a 1 : 1 : 1 ratio. The mineral is represented as:**

(A) $CaCuSi_3O_{10} \cdot H_2O$
(B) $CaCuSi_3O_{10} \cdot 2H_2O$
(C) $Ca_2Cu_2Si_3O_{10} \cdot 2H_2O$
(D) None of these

Sol. **(C)**

It contains $Si_3O_{10}^{8-}$ units.

Q33 **Zeolites are minerals of the composition:**

$$M_{\left(\frac{x}{n}\right)}\left[\left(AlO_2\right)_x\left(SiO_2\right)_y\right].zH_2O$$

Where 'n' is the charge on the metal cation. These structures contain various rings which result in formation of a fairly open structure containing channels and cavities at molecular level (2 to 11 Å) in diameter. They may be used to trap molecules of various sizes. This makes them very useful as "Selective adsorbents" or as "molecular sieves". In the structure of zeolites:
(A) Each Al atom is attached to 'p' oxygen atoms directly.
(B) Each Si atom is attached to 'q' oxygen atoms directly.
(C) Each Al atom is attached to 'r' Si atoms directly.
Find (p + q + r).

Sol. **(8)**

p =4, q = 4, r = 0

Tin and its Compounds

Important ore: Cassiterite (tin stone) SnO_2

Compounds of Tin

(i) Stannic oxide, SnO_2:

Preparation:

(1) By heating tin strongly in air.

$$Sn + O_2 \longrightarrow SnO_2$$

(2) By hot concentrated nitric acid
Tin treated with hot concentrated nitric acid will give meta stannic acid which give stannic oxide on heating.



$$Sn + 4HNO_3(conc.) \longrightarrow H_2SnO_3 + 4NO_2 + H_2O$$

$$H_2SnO_3 \xrightarrow{\Delta} SnO_2 + H_2O$$

Properties:

- It is a white amphoteric solid.
- It is insoluble in water(H_2O)
- Reaction with NaOH/KOH

$Sn + 2NaOH + H_2O \rightarrow Na_2SnO_3 + 2H_2$

In absence of air Na_2SnO_2 form and as it contacts with air it converts into Na_2SnO_3.

Uses:

- It is used Tiles industry and pottery.
- It is also used to making powder for polishing purpose

Q34 **Sn reacts with HNO_3 under different conditions. What are various products?**

Sol. $4Sn + \text{dil. } 10HNO_3 \rightarrow 4Sn(NO_3)_2 + NH_4NO_3 + 3H_2O$

$Sn + 4HNO_3 \rightarrow H_2SnO_3 + H_2O + NO_2$

(ii) Stannous oxide, SnO:

Preparation:

(1) By heating stannous oxalate

$$SnC_2O_4 \xrightarrow{\Delta} SnO + CO + CO_2$$

It observing that if CO is formed then its significant of stannous oxide

(2) By heating SnO in presence of air.

$$2SnO + O_2 \longrightarrow 2SnO_2$$

Properties:

- It is grey amphoteric solid.
- By treating with base (NaOH) it will give Na_2SnO_2.

Tin(II) Oxide

SnO



Q35 **Tin shields in severe winter, fall to powder.**

Sol. Due to Higher melting point and ionic character

(iii) Stannous sulphide, SnS:

- It is insoluble in water but soluble only in hot conc. HCl .
- Its get converted into ammonium thiostannate $(NH_4)_2SnS_3$. While treating with yellow ammonium polysulphide.

(iv) Stannous chloride, $SnCl_2$:

Preparation:

(1) By concentrated HCl

$$Sn(s) + HCl(Conc.) \longrightarrow SnCl_2(aq) + H_2(g)$$

(2) By concentrating this above resulting solution, it will give crystals of $SnCl_2.2H_2O$. Simply it is heated, basic tin chloride is achieved.

$$SnCl_2.2H_2O(s) \xrightarrow{\Delta} Sn(OH)Cl + HCl + H_2O$$

To obtain anhydrous $SnCl_2$, heat Sn in dry HCl gas.

$$Sn + 2HCl \xrightarrow{\Delta} SnCl_2 + H_2$$

Tin(II) Chloride
$SnCl_2$





Properties:

(a) A Reaction with HCl
$SnCl_2 + 2HCl \rightarrow HSnCl_3 + HCl \rightarrow H_2SnCl_4$

(b) Reduction of $SnCl_2$
It is good reducing agent and It reduce ferric, cupric, mercury chlorides etc.
$Sn^{+2} + 2F^{+3} \rightarrow 2F^{+2} + Sn^{+4}$
$2Cu^{+2} + Sn^{+2} \rightarrow 2Cu^{+} + Sn^{+4}$
$Hg^{+2} + Sn^{+2} \rightarrow Hg + Sn^{+4}$

(c) It combines with I2 and form $SnCl_2I_2$

(v) Stannic chloride, $SnCl_4$:

Preparation:

(a) It is achieved by the action of Cl_2 on molten Sn.

$$Sn + 2Cl_2 \longrightarrow SnCl_4$$

(b) By treating with dilute $HgCl_2$

$$Sn + 2HgCl_2 \longrightarrow SnCl_4 + 2Hg$$

Properties:

(a) $SnCl_4.5H_2O$ with excess water it is hydrolysed to form basic chloride and ultimately stannic acid (H_2SnO_4).

$$SnCl_4 + H_2O \longrightarrow Sn(OH)Cl + HCl$$

$$Sn(OH)Cl + 3H_2O \longrightarrow Sn(OH)_4 \text{ or } H_2SnO_4 + 3HCl$$

Uses:

- It is used to form some important double salts with NH_3, N_2O, PCl_5

e.g., $SnCl_4.4NH_3$.

- It is used in formation of tinning agent

Q36 **$SnCl_2$ is solid but $SnCl_4$ is liquid.**

Sol. $SnCl_2$ has ionic character and thus has higher m.p.t and exists as solid. On the hand $SnCl_4$ involves sp^3 hybridisation in Sn atom leading to covalent character and thus is liquid.

Lead

Some important ores

Galena PbS (Main)

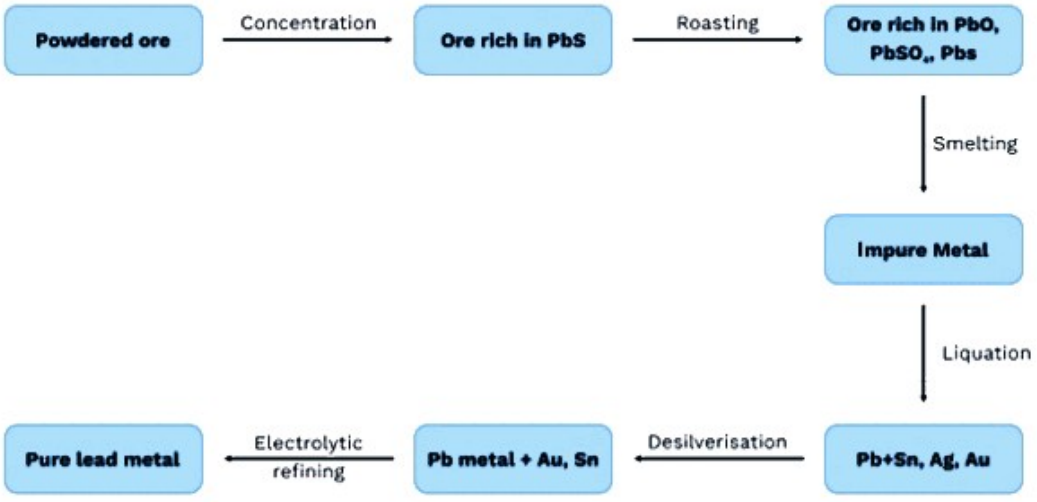
Cerussite $PbCO_3$

Anglesite $PbSO_4$

Lanarkite $PbO.PbSO_4$



Extraction from Galena



Compounds of Lead

(i) Lead oxide (Litharge),PbO :

Lead(II) Oxide
PbO



Preparation:

Laboratory prepration

$$2Pb(NO_3)_2 \xrightarrow{\Delta} 2PbO + 4NO_2 + O_2$$

Lead oxide found in two different forms
First one is yellow which and under fused cooled condition it convert into reddish yellow litharge.

Properties:

It is insoluble in water and only soluble in NaOH

$$PbO + 2NaOH \longrightarrow Na_2PbO_2 + H_2O$$

It is amphoteric solid.

It can be reduced with several reducing agents (etc.) to lead(Pb).

Uses:

- It is used in glass industry
- It is used in paints and petroleum industry.

Lead(IV) Oxide
PbO_2



(ii) Lead dioxide, PbO_2:

Preparation:

By heating of Pb_3O_4 with nitric acid

$$Pb_3O_4 + 4HNO_3 \xrightarrow{\Delta} 2Pb(NO_3)_2 + 2H_2O + PbO_2$$

Properties:

It is amphoteric in nature and dissolves in NaOH to form sodium plumbate.

$$2NaOH + PbO_2 \longrightarrow Na_2PbO_3 + H_2O$$

It is good and powerful oxidising agent.

Examples:

$$PbO_2 + 4HCl \xrightarrow{Warm} PbCl_4 + H_2O$$

- It is insoluble in water as well as in nitric acid.
- Its nature is amphoteric and is utilized in lead storage batteries and in safety matches.

Q37 **Explain why the reactions with HCl and SO_2 show the oxidising nature of PbO_2?**

Sol. $PbO_2 + 4HCl \rightarrow PbCl_2 + Cl_2 + H_2O$

$PbO_2 + SO_2 \rightarrow PbS + SO_4$

There PbO_2 (Pb is present in (+4) oxidation State) it cant increase further it's oxidation state. Thus it reduce.

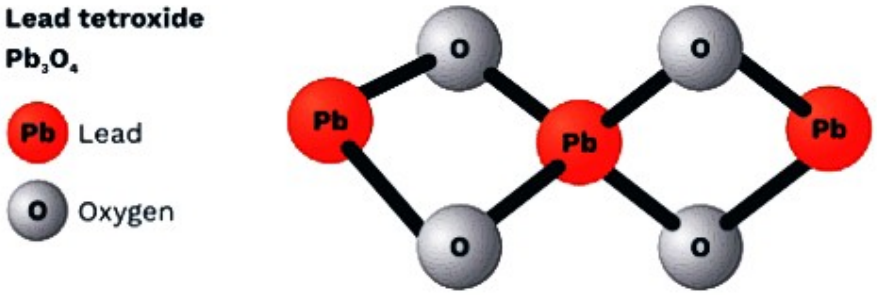
(iii) Sindhur or Red lead, Pb_3O_4:

Preparation:

By heating PbO in air above 673K.

$$6PbO + O_2 \xrightarrow{673K} 2Pb_3O_4$$

Lead tetroxide
Pb_3O_4



Properties:

It is a red crystalline solid unsolvable in water.

It is a mixed oxide of $PbO_2 + 2PbO$ and reacts with HNO_3 to form $Pb(NO_3)_2$ and PbO_2.

$2PbO.PbO_2 + 4HNO_3 \rightarrow Pb(NO_3)_2 + 2H_2O + PbO_2$

It is a good and strong oxidising agent.

Example:

$Pb_3O_4 + 8HCl \longrightarrow 3PbCl_2 + 4H_2O + Cl_2$

$2Pb_3O_4 + 6H_2SO_4 \longrightarrow 6PbSO_4 + 6H_2O + O_2$

Uses:

- It is used as a protection layer on rust as a paint.
- It is used in glass industry

Q38 **Predict the result of reacting Pb_3O_4 with concentrated hydrochloric acid.**

Sol. $Pb_3O_4 + \text{con. } 8HCl \rightarrow 3PbCl_2 + 4H_2O + Cl_2$.

Q39 **Predict the result of reacting Pb_3O_4 with conc. H_2SO_4.**

Sol. $2Pb_3O_4 + \text{conc. } 6H_2SO_4 \rightarrow 6PbSO_4 + 6H_2O + O_2$

Q40 **PbO does not dissolve in H_2SO_4 while SnO is soluble in H_2SO_4.**

Sol. $PbO + H_2SO_4 \rightarrow PbSO_4 + H_2O$

(insoluble)

$SnO + H_2SO_4 \rightarrow SnSO_4 + H_2O$

(Soluble)

(iv) Lead chloride, $PbCl_2$:

Preparation:

(A) It can be made by treating a salt of lead with dil. HCl.

$$Pb(NO_3)_2 + 2HCl \longrightarrow PbCl_2 + 2HNO_3$$

Lead(II) Chloride
$PbCl_2$



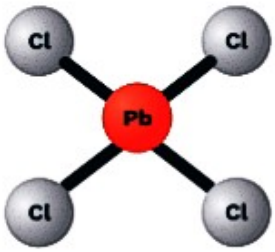


Properties:

Reaction with calcium hydroxide

$$PbCl_2 + Ca(OH)_2 \longrightarrow Pb(OH)Cl + CaO + HCl$$

(v) Lead tetrachloride, $PbCl_4$:



Preparation:

By heating of PbO_2 with conc. HCl.

$$PbCl_4 \xrightarrow{373K} PbCl_2 + Cl_2$$

It exist in form of complex hexachloroplumbate (IV) ion.

$$PbCl_4 + 2HCl \longrightarrow H_2[PbCl_6]$$

Q41 **(i) A white precipitate (B) is formed when a mineral (A) is boiled with Na_2CO_3 solution.**
(ii) The precipitate is filtered and the filtrate contains two compounds (C) and (D). The compound(C) is removed by crystallization and when CO_2 is passed through the mother liquor left, (D) changes to (C).
(iii) The compound (C) on strong heating gives two compounds (D) and (E).
(iv) (E) on heating with cobalt oxide produces blue coloured substance (F).
Identify (A) and (F) and give chemical reactions for the steps (i) to (iv).

Sol. **(i)** $Ca_2B_6O_{11} + 2Na_2CO_3(aq) \xrightarrow{\text{Boiled}} 2CaCO_3 + Na_2B_4O_7 + 2NaBO_2$

(A) (B) (C) (D)

Calemanite — Insoluble — Soluble

(C) is borax (crystallized)

(ii) The mother liquer consisting sodium metal borate is treated with CO_2.

$4NaBO_3 + CO_2 \rightarrow Na_2B_4O_7 + Na_2CO_3$

(C)

(iii) $Na_2B_4O_7 10H_2O \xrightarrow{\Delta} Na_2B_4O_7 + 10H_2O$

anhydrous

$Na_2B_4O_7 \xrightarrow{\Delta} 2NaBO_2 + B_2O_3$ (E)

(iv) $CoO + B_2O_3 \xrightarrow{\Delta} Co(BO_2)_2$

Cobalt metaborate

Blue coloured (F)

Q42 **A certain salt (X) gives the following tests.**
(i) Its aqueous solution is alkaline to litmus.
(ii) On strongly heating it swells to give glassy material.
(iii) When concentrated H_2SO_4 is added to a hot concentrated solution of (X), white crystals of a weak acid separate out. Identify (X) and write down the chemical equations for reaction at steps (i), (ii) and (iii).

Sol. Certain salt X gives fests-

(i) aqueous solution is alkaline to litmus.

$Na_2B_4O_7 + 7H_2O \rightarrow 2NaOH + 4H_3BO_3$

Strong alkali — Weak acid

(ii) On strong heating it swells to give glassy material

$$Na_2B_4O_7.10H_2O \xrightarrow{60°C} Na_2B_4O_7 5H_2O$$

$$\downarrow \Delta$$

(Anly drous swollen white dark) $Na_2B_4O_7$

$$\downarrow \Delta$$

$$2NaBO_2 + B_2O_3$$

Glassy mass

(iii) Reaction with concentrated H_2SO_4

$$Na_2B_4O_7 + H_2SO_4 + 5H_2O \rightarrow Na_2SO_4 + 4H_3BO_3$$

Q43 **An Inorganic Lewis acid (X) shows the following reactions:**
(i) It fumes in moist air.
(ii) The intensity of fumes increases when a rod dipped in NH_4OH is brought near it.
(iii) An acidic solution of (X) on addition of NH_4Cl and NH_4OH gives a precipitate which dissolves in NaOH solution.
(iv) An acidic solution of (X) does not give precipitate with H_2S. Identify (X) and give chemical equations for reactions at steps (i) to (iii).

Sol. Inorganic lewis acid X = $AlCl_3$

(i) Fumes is moist air

$$Al_2Cl_6 + 6H_2O \rightarrow 2Al(OH)_3 + 6HCl\uparrow$$

(evolution of fume)

(ii) When a rod dipped in NH_4OH is brought near to Ct, arise intensity of times. because NH_4OH is basic in nature

(iii) Acidic solution of X on addition of NH_4Cl and NH_4Cl and NH_4OH gives precipitate Aqueous solution of $AlCl_3$

$$AlCl_3 + 3H_2O \longrightarrow Al(OH)_3 + 3HCl$$

\+ white precipitin

(Precipitate dissolve in excess NaOH) NaOH (excess)

$$\downarrow$$

$$NaAlO_2 + 2H_2O$$

Sodium meta aluminate

Q44 **Certain metallic oxides are reduced with aluminium rather than with carbon. Why?**

Sol. Al have low ionization energy. Thus it can gives easily electron to other So, behave as strong reducing agent. Carbon have high ionization energy thus it can't gives easily electron.

Q45 **How will you prepare the following -**
(a) Tin metal from cassiterite.
(b) Red lead from litharge.
(c) Stannous oxide from stannous chloride.
(d) Crystalline silicon from silicon tetrachloride.
(e) Chrome yellow from litharge.

Sol. **(a)** After roasting of cassiterite are about one fifth of its mas of powdered anthracite and a little lime is mixed. The mixture is heated in the reverberatory furnance.

$SnO_2 + 2C \rightarrow Sn + CO$

(b) $6PbO + O_2 \rightarrow 2Pb_3O_4$

(c) NaOH is added to stannous chloride. The stannous hydroxide precipitated is heated in atmosphere of CO_2.

$SnCl_2 + 2NaOH \rightarrow Sn(OH)_2 + 2NaCl$

$Sn(OH)_2 \xrightarrow{\text{Heat } Cr_2} SnO + H_2O$

(d) By passing current of silicon tetrachloride over molten aluminium.

$3SiCl_4 + 4Al \rightarrow 3Si + 4AlCl_3$

(e) $PbO + CH_3COOH \rightarrow (CH_3COO)_2Pb + H_2O$

$(CH_3COO)_2Pb + K_2CrO_4 \rightarrow PbCrO_4 + 2CH_3COOK$

chrome yellow

Q46 **Identify (A), (B), (C) , (D) and (E) :**

(a) Si $\xrightarrow[\text{Conc., } \Delta]{HNO_3/HF}$ (A) $\xrightarrow{HF}$ (B)

(A) $\xrightarrow{H_2O}$ (B) + (C)

(b) Graphite $\xrightarrow{\text{hot conc. } HNO_3}$ **(D)**

(c) $\mathbf{Na_2CO_3}$ **(fused) + Si** $\rightarrow$ **(E)**

Sol.

(a) $Si \xrightarrow[\Delta]{Con.HNO_3/HF} (A)\ SiF_4 \xrightarrow{HF} (B)\ H_2SiF_6$

$\downarrow H_2O$

$(B)\ H_2SiF_6 + (C)\ H_4SiO_4$

(b) Graphite $\xrightarrow{Hot.con.HNO_3}$ (D) $C_6(COOH)_6$

Mellitic Acid

(c) $Na_2CO_3(fused) + Si \rightarrow (E)\ Na_2SiO_3 + C$

Q47 **How will you prepare the following -**

(a) Litharge from red lead.

(b) Silanes from silicon.

(c) Tetraethyl lead from sodium lead alloy.

Sol.

(a) $2Pb_3O_4 \rightarrow 6PbO + O_2$

(b) lead is converted in to white lead by dutch process or electrolytic process.

(c) $2Mg + Si \rightarrow MgSi \xrightarrow{dil.\ HCl}$ A mixture of Silicon Hydrides

Silanes

Q48 **What happens when -**

(a) Dilute nitric acid is slowly reacted with tin.

(b) Carbon and steam are heated.

(c) Iodine is added to stannous chloride.

(d) Tin is treated with concentrated nitric acid.

(e) NaOH solution is added to $SnCl_2$ solution.

(f) Stannous chloride reacts with ferric chloride.

(g) Nitrobenzene is treated with acidified stannous chloride.

(h) Silicon is fused with soda ash.

Sol.

(a) $4Sn + 10\ dil.\ HNO_3 \rightarrow 4Sn(NO_3)_2 + NH_4NO_3 + 3H_2O$

(b) $C + H_2O \rightarrow CO + H_2$

Water gas

(c) $SnCl_2 + 2I_2 \rightarrow SnI_4 + Cl_2$

(d) $Sn + 4\ Con.\ HNO_3 \rightarrow H_2SnO_3 + 2H_2O + 4NO_2$

m-Stannic Acid

(e) $SnCl_2 + 2NaOH \rightarrow Sn(OH)_2 + 2NaCl$ (white ppt)

(f) $2SnCl_2 + FeCl_3 \rightarrow 2FeCl_2 + SnCl_4$

(g) $C_6H_5NO_2 + 6HCl + SnCl_2 \rightarrow C_6H_5NH_2 + 2H_2O + 3SnCl_4$

Aniline

(h) $4NaOH + SiC + 2O_2 \rightarrow Na_2SiO_2 + Na_2CO_3 + 2H_2O$

Q49 **Solution of CO_3^{2-} (say Na_2CO_3) is alkaline.**

Sol. Alkali metal (IA), Highly electropositive thus, it is alkaline in nature.

$CO_3^{2-} + 2H_2O \rightleftharpoons H_2CO_3 + 2OH^-$

alkaline

Q50 **Carbon never shows coordination number greater than four while other members of carbon family can show co-ordination number greater than four.**

Sol. Due to absence of 'd' orbital.

Q51 **Sn^{2+} and Fe^{3+} cannot coexist in the same solution.**

Sol. Sn^{2+} K_{sp} (solubility product value) is low.

But Fe^{3+} K_{sp} (solubility product) is High Thus, cant exists in same solution.

Q52 **Alkali cannot be stored in a Sn or Pb vessel.**

Sol. Alkali can't be store in Sn or Pb vessel because alkali are more reducing agent than Sn and Pb.

Miscellaneous Illustrations (Bororn Family)

Q53 **The stable oxidation state of Tl is +1 hence its hydroxide is similar to -**

(A) $Ca(OH)_2$ **(B) $Al(OH)_3$** **(C) $Mg(OH)_2$** **(D) NaOH**

Sol. **(D)**

Tl^{+1} stable i.e. Na^+ OH^-

Q54 **Which is true for an element R present in 13th group of the periodic table ?**

(A) It is gas at room temperature **(B) It has oxidation state of + 4**

(C) It forms R_2O_3 **(D) It forms RX_2**

Sol. **(C)**

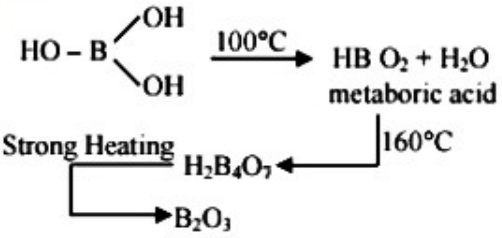
Group 13 elements are in +3 and +1 oxidation state.

thus they form R_2O_3 Type oxide. Ex. Al_2O_3

Q55 **Boric acid heated to red hot gives -**

(A) HBO_2 **(B) H_2BO_2** **(C) B_2O_3** **(D) Borax**

Sol. **(C)**



Q56 **The solution of borax in water is.....**

Sol. Alkaline

Solution of Borax is alkaline in nature due to Hydrolysis

$Na_2B_4O_7 + 7H_2O \rightarrow 2NaOH + 4H_3BO_3$

Q57 **In the following reaction -**

$B(OH)_3 + H_2O \longrightarrow [B(OH)_4]^- + H^+$

(A) $B(OH)_3$ is a Lewis acid **(B) $B(OH)_3$ is a Lewis base**

(C) $B(OH)_3$ is amphoteric **(D) None is correct**

Sol. **(A)**

$B(OH)_3 + H_2O \longrightarrow [B(OH)_4]^- + H^+$

$\bar{e}$ acceptance ¾→ lewis acid.

Q58 **B_4C is also known as -**

(A) Cast steel **(B) Carbo boron**

(C) Boron carbide **(D) Carbon steel with boron impurity**

Sol. **(C)**

B_4C = Boron carbide

Sol. $SnCl_2$ has ionic character and thus has higher m.p.t and exists as sold. On the hand $SnCl_4$ involves sp^3 hybridisation in Sn atom leading to covalent character and thus is liquid.

Q64 **Carborundum is -**

(A) SiO2 **(B) SiF4** **(D) Si3N4** **(D) SiC**

Sol. **(D)**

SiD = Carborundum

Q65 **The Basic building unit of silicates is -**

(A) SiO_6^{2-} **(B)** SiO_4^{4-} **(C)** SiO_2^{6-} **(D)** SO_6^{2+}

Sol. **(B)**

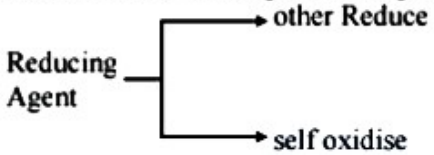
Forms various silicates

Q66 **Silanes are strong ..... agents**

(A) Oxidising **(B) Reducing**

(C) Dehydrating **(D) Bleaching**

Sol. Silicones are strong reducing agent reducing



Q67 **In SiO^-, silicon is in -**

(A) sp – hybrid state **(B) sp^2 – hybrid state**

(C) sp^3 – hybrid state **(D) dsp^3 – hybrid state**

Sol. **(C)**

sp^3 – hybrid state.[gaint molecule formation by tetrahedral Si atom]

Q68 **The structures of quartz, mica, asbestos have the common basic unit of**

(A) $(SiO_4)^{4-}$ **(B)** $(SiO_3)^{2-}$ **(C)** $(SiO_4)^{2-}$ **(D)** SiO_2

Sol. **(A)**

Quart, mica, asbestos have common unit $(SiO_4)^{4-}$

Q69 **SnO_2 is**
(A) Acidic **(B) Alkaline** **(C) Amphoteric** **(D) Neutral**

Sol. **(C)**

SnO_2 is amphoteric

Q70 **Tin (II) chloride is used**
(A) As a mordant in dyeing **(B) As a reducing agent**
(C) In the preparation of colloidal gold **(D) All of these**

Sol. **(D)**

$Sn^{2+} \longrightarrow Sn^{4+}$

Ability to form colloidal

Q71 **Which one of the following chlorides is both a covalent liquid and rapidly hydrolyzed by water**
(A) $PbCl_4$ **(B) PCl_5** **(C) CCl_4** **(D) $PbCl_2$**

Sol. **(A)**

Pb^{4+} hence can by hydrolyzed easily

Q72 **PbO_2 is**
(A) Acidic **(B) Basic** **(C) Neutral** **(D) Amphoteric**

Sol. **(D)**

PbO_2 amphoteric

Q73 **When chlorine is passed through molten tin, the product obtained is**
(A) $SnCl_2$ **(B) $[SnCl_4]^{2-}$** **(C) $[SnCl_6]^{2-}$** **(D) $SnCl_4$**

Sol. **(D)**

$Sn + 2Cl_2 \longrightarrow SnCl_4$

Molten tin

Q74 **Choose the incorrect option**

(A) Tendency of form ionic compound increases from B to Tl

(B) Boron forms only covalent compound

(C) Thallium forms only ionic compounds

(D) Group 13 elements have less tendency to form complexes than the s-block elements

Sol. **(D)**

Group 13 element have more tendency to form complex than s-block element

$\longrightarrow$ due to their small size L more effective nuclear charge as well as vacant orbitals to accept electrons.

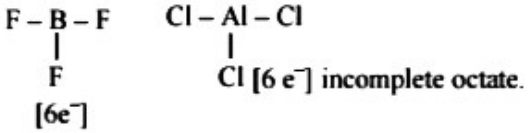
Q75 **Boron compounds behave as Lewis acids because of their -**

(A) Acidic nature **(B) Covalent nature**

(C) Electron deficient character **(D) Ionising property**

Sol. **(C)**

Boron compound behave as lewis acid character because in Boron group it valence shell electron is three. Thus it's octet is incomplete.



Q76 **On small scale, boron is best isolated by reducing boron trioxide with -**

(A) Metallic zinc **(B) Metallic aluminium**

(C) Metallic copper **(D) Metallic magnesium**

Sol. **(D)**

$B_2O_3 + 3Mg \xrightarrow{1100°C} 2B + 3MgO$

Q77 **Boron reacts with nitric acid to form**

(A) Sodium borate, H_2 **(B) Boric acid**

(C) Diborane **(D) Borax**

Sol. **(B)**

$B + 3HNO_3 \rightarrow \underset{\text{Boric acid}}{H_3BO_3}$

Boron remains unaffected by. Non oxidising acids.

Q78 **In the electrolysis of method of boron extraction the cathode is made of**

(A) Carbon **(B) Boric anhydride** **(C) Mg** **(D) Iron rod**

Sol. **(D)**

Electrolysis of fused B_2O_3 + MgO + MgF temp. = 1373 K.

anode = carbon & cathode = iron rod

Q79 **The shape of jeweller's borax is**

(A) Prismatic **(B) Octahedral**

(C) Triangular **(D) Tetrahedral**

Sol. **(B)**

$Na_2B_4O_7.5H_2O$ = Jeweller's borax

Q80 **H_3BO_3 on neutralization with.....gives borax.**

(A) B_2O_3 **(B) $CaCO_3$** **(C) Na_2CO_3** **(D) NaOH**

Sol. **(C)**

$4H_3BO_3 + Na_2CO_3 \rightarrow Na_2B_4O_7 + 6H_2O + CO_2$ Borax

Q81 **The power of halides of boron to act as lewis acids decreases in the order**

(A) $BF_3 > BCl_3 > BBr_3$ **(B) $BBr_3 > BCl_3 > BF_3$**

(C) $BCl_3 > BF_3 > BBr_3$ **(D) $BCl_3 > BBr_3 > BF_3$**

Sol. **(B)**

Order of lewis acid strength of boron. halides is $BI_3 > BBr_3 > BCl_3 > BF_3$

Q82 Which one of the following statement regarding BF_3 is not correct –

(A) It is a lewis acid

(B) It is an ionic compound

(C) It is an electron deficient compound

(D) It forms adducts

Sol. (B)